ENHANCING LOW-RESOURCE LANGUAGE PROCESSING: FINE-TUNING MARIANMT FOR ILOCANO-TAGALOG NEURAL MACHINE TRANSLATION

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DANNIEL PAUL B. MOLLENO
DANIA MARIE BAUTISTA
REBECCA B. GOMEZ

MARIBEL V. CARDONA

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APPROVAL SHEET

The thesis entitled **"ENHANCING LOW-RESOURCE LANGUAGE PROCESSING: FINE-TUNING MARIANMT FOR ILOCANO-TAGALOG NEURAL MACHINE TRANSLATION"** prepared and submitted by **Dania Marie Bautista, Rebecca B. Gomez,** and **Danniel Paul B. Molleno** in partial fulfillment of the requirements for the degree Bachelor of Science in Computer Science specialized in Intelligent Systems, is hereby recommended for approval and acceptance.

MARIBEL V. CARDONA
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of ___.

GENE MARCK B. CATEDRILA
Member

CAROLINA R. JOVAL
Member

JOCELYN O. PADALLAN
Member

JEFFERSON L. LERIOS
Member

Accepted as partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science specialized in Intelligent Systems.

LOYD S. ECHALAR, DIT
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, EdD.
Chairperson, Research and Development Office

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

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D.P.B.M.

ABSTRACT

A controlled neural machine translation framework for Ilocano-Tagalog is presented to address data scarcity in low-resource Philippine languages. Using a MarianMT-based architecture, two models were fine-tuned: a non-augmented baseline and a mask-based augmented model. A parallel corpus of 10,038 high-quality sentence pairs was derived from 42,273 raw entries through cleaning and alignment. Mask-based augmentation expanded the training set from 8,130 to 32,520 pairs, improving exposure to diverse lexical and syntactic patterns. Translation quality was evaluated using BLEU, ChrF, METEOR, human assessment, and paired bootstrap resampling. The augmented model outperformed the baseline (BLEU = 88.24, ChrF = 92.86, METEOR = 0.9284), with statistically significant improvements ($p < 0.001$) and fewer errors in human evaluation. Although the absolute BLEU, ChrF, and METEOR scores are high, these values reflect the controlled nature of the dataset, which is drawn from selected domains with limited syntactic variability and minimal idiomatic divergence between Ilocano and Tagalog. Accordingly, the primary focus is the relative performance improvement between augmented and non-augmented models rather than the absolute magnitude of the scores. Results demonstrate that mask-based augmentation enhances translation accuracy and consistency, highlighting its potential for low-resource language NMT. Future work may explore additional data sources, augmentation techniques, and multilingual transfer.

Keywords – *Ilocano-Tagalog translation, Neural Machine Translation, MarianMT, Low-resource languages, Mask-based Data Augmentation, Bootstrap Resampling*

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DEFINITION OF TERMS

This section presents definitions of the key terms used in the study, both technical and operational, to ensure clarity and unambiguity. The technical definitions provide the common meaning of each term; the operational definitions explain how each term will be defined within the study in the context of Ilocano-Tagalog translation.

Augmentation	A data preparation technique that increases training data variability by modifying existing text samples. In this study, a mask-based augmentation approach is used to improve model generalization and translation performance for low-resource Ilocano-Tagalog neural machine translation.
Adequacy	The extent to which the translated sentence conveys all essential information present in the source text. In this study, adequacy is assessed by whether the Tagalog translation sufficiently captures the whole meaning of the Ilocano source sentence, even when differences in wording or sentence structure occur.
Bilingual Evaluation Understudy (BLEU) Score	An automatic evaluation metric for machine translation quality based on the precision of overlapping n-grams between a machine translation and one or more human reference translations.
ChrF (Character n-gram F-score)	An automatic machine translation evaluation metric that measures similarity between generated and reference translations using character-level n-gram precision and recall. In this study, ChrF is used to better capture

	morphological variations in Ilocano-Tagalog translations.
Cohen's Kappa (κ)	A statistical measure of inter-annotator agreement that accounts for agreement occurring by chance. Cohen's Kappa is used in this study to quantify the level of consistency between human evaluators during Likert-scale human evaluation of translation quality.
Dataset	An organized set of Ilocano-Tagalog parallel sentence pairs amounting to some 42,273. Herein, the data set shall serve as the primary data for training and cross-validation, and as the test set for the MarianMT model.
Error Analysis	The systematic examination of translation errors to identify patterns and weaknesses in machine translation outputs. In this study, error analysis is used to assess errors in Ilocano-Tagalog translations qualitatively.
Evaluate	An assessment process was conducted to measure the translation quality of MarianMT. In this study, the evaluation was performed using automatic metrics, specifically the BLEU score.
Faithfulness	The degree to which a translated output accurately preserves the original meaning, intent, and content of the source text without introducing omissions, additions, or distortions. In this study, faithfulness refers to how correctly the Tagalog translation reflects the meaning of the Ilocano source sentence, as evaluated through human judgment.

Fine-Tuning	A process of adjusting a pre-trained language model to a specific task or dataset using a smaller, specialized corpus. Fine-tuning allows models like MarianMT to adapt to language pairs or domains, improving performance without retraining from scratch.
Fluency	The degree to which the translated output is grammatically correct, natural-sounding, and consistent with the linguistic norms of the target language. In this study, fluency refers to how smooth, clear, and readable the Tagalog translation is, independent of its accuracy to the source text.
Generalization	The ability of a machine learning model to perform well on unseen data. In this study, generalization refers to the model's effectiveness in translating Ilocano-Tagalog test data.
Ilocano-Tagalog Parallel Corpus	A collection of sentence pairs in both Tagalog and Ilocano languages, used to train and evaluate the machine translation model.
Library	A set of machine learning frameworks and NLP toolkits that offer built-in functions for tokenization, training, and evaluation. A library is used in this study to implement and fine-tune the MarianMT model.
Low-Resource Language	A language that lacks large-scale digital corpora, parallel datasets, and computational linguistic resources can make it challenging to develop high-quality machine translation or natural language processing systems. In this study, Ilocano is considered a low-resource language.

MarianMT	An open-source neural machine translation framework optimized for speed and efficiency, used in this study to fine-tune a translation model for Tagalog and Ilocano.
Masking	A preprocessing technique was applied in which, during training, specific tokens in a sentence were concealed or replaced. This masking procedure enables MarianMT to infer the missing words, thereby enhancing the contextual understanding of the target languages, Tagalog and Ilocano, throughout the translation process.
METEOR (Metric for Evaluation of Translation with Explicit ORdering)	An automatic evaluation metric that measures translation quality using unigram matching with consideration of precision, recall, stemming, synonyms, and word order. In this study, METEOR is used to assess semantic similarity between translations.
Neural Machine Translation (NMT)	An approach to automatic language translation that uses deep learning models, particularly neural networks, to predict and generate translations. NMT systems generally outperform traditional statistical machine translation methods.
Overfitting	A condition where a model performs well on training data but poorly on unseen data. In this study, overfitting is monitored by comparing training and validation performance.
Paired Bootstrap Resampling	A statistical significance testing method used in machine translation evaluation to determine whether the performance difference between two translation

	systems is statistically meaningful. The method repeatedly resamples the same set of test sentences with replacement and compares the evaluation scores of two systems for each resampled subset. Because the same sentence samples are used for both systems, the test controls for sentence-level variability.
Parallel Pairs	A collection of aligned Ilocano and Tagalog sentences that have the same meaning. The parallel pairs organize the data to provide input-output examples directly for training MarianMT.
SentencePiece	An unsupervised text tokenizer and detokenizer are primarily used for neural network-based text generation tasks and help prepare text by creating subword units suitable for training models.
Test Set	The final subset of the dataset, approximately 10%, was withheld for testing MarianMT after training. For this study, the test set provides an unbiased estimate of the translation accuracy and fluency of the MarianMT model.
Tokenization	The process of splitting text into smaller units, such as words, sub-words, or characters, makes it more manageable for neural networks to process during machine translation or natural language processing.
Training Set	A subset of the dataset, constituting approximately 90% of the total, is allocated for training MarianMT in translation patterns, grammar, and

	vocabulary. In this study, the training set enables the model to learn the mappings between Ilocano and Tagalog.
Validation Set	A smaller subset of the dataset, approximately 10% of the training set, was used while training to tune and optimize hyperparameters, adapt to learning rates, and reduce overfitting. The validation set, in this study, is to show that MarianMT generalizes beyond the training set data.
Zipf's Law	A linguistic principle stating that the frequency of words in a language exhibits a bidirectional relationship with respect to rank. In this paper, this principle highlights the natural bias inherent to both common and rare words in the Ilocano-Tagalog dataset, and thereby to translation performance in low-resource circumstances.

CHAPTER 1

INTRODUCTION

Background of the Study

The digital revolution has fundamentally reshaped the function of language. Once primarily a means of interpersonal communication, language has increasingly become a gateway to information access, civic participation, and engagement in public debate. In contemporary digital environments, the language a person speaks can determine whether they can access essential services, participate meaningfully in online discourse, and benefit from technological innovation. In multilingual countries such as the Philippines, this relationship between language and digital participation has significant social implications (Roche, 2023).

The Philippines is among the most linguistically diverse countries in the world, with more than 180 languages spoken across over 7,000 islands (Eberhard, Simons, and Fennig, 2020; SIL International, n.d.). Despite this diversity, digital language technologies have historically prioritized only a small number of dominant languages, particularly English and Tagalog. Tagalog, spoken by tens of millions of Filipinos, has been incorporated into numerous multilingual natural language processing models and translation standards and has received relatively greater attention in open-source language technology development (Liu et al., 2020; Tiedemann, 2020). As a result, Tagalog has achieved a level of digital visibility unmatched by many other Philippine languages.

In contrast, Ilocano, despite being spoken by millions of Filipinos and actively used in everyday communication, remains underrepresented in the digital language technology landscape. Compared to dominant languages such as Tagalog and English, Ilocano has far fewer computational tools, annotated corpora, and translation resources available (Casil, 2018). Some existing data also reflect natural code-switching practices, particularly the mixing of English and Tagalog elements, which further complicates the development of clean and well-aligned Ilocano language resources.

This underrepresentation contributes directly to language-based inequality. Ilocano-speaking communities often encounter difficulties in accessing digital content, public services, health information, and educational materials in their native language. These limitations restrict meaningful participation in digital society and reinforce existing regional and social disparities (Joshi et al., 2020). Moreover, recent studies highlight a gradual shift among younger speakers toward increased use of Tagalog and English, especially in digital and educational contexts. While Ilocano remains actively spoken in many rural areas, the continued lack of technological support may weaken intergenerational language transmission and accelerate language attrition over time (Lachica & Albert, 2024).

Global initiatives such as UNESCO's International Decade of Indigenous Languages and Meta AI's No Language Left Behind campaign underscore the growing recognition of the need for inclusive language technologies to preserve linguistic diversity and bridge the digital divide (Meta AI, n.d.; UNESCO, 2021). However, despite these efforts, many multilingual machine translation systems

continue to rely on large-scale multilingual datasets and substantial computational resources. These requirements limit their effectiveness for low-resource languages, particularly in contexts where available corpora are small, unevenly distributed, or domain-specific, as is often the case for Philippine languages (Ranathunga et al., 2021).

Consequently, pre-trained multilingual translation models often underperform on languages such as Ilocano. The lack of sufficient language-specific data hinders the model's ability to capture morphological variation, lexical diversity, and context-dependent meaning. These limitations highlight the need for targeted adaptation strategies that allow existing models to be specialized for low-resource language pairs rather than relying solely on broad multilingual training.

To address these challenges, this study focuses on fine-tuning the MarianMT framework for Ilocano and Tagalog translation. Fine-tuning involves continuing the training of a pre-trained neural machine translation model on a smaller, language-specific dataset, allowing the model to adapt more effectively to the linguistic characteristics of a particular language pair. This approach is especially suitable for low-resource settings, enabling efficient use of existing data while avoiding the high computational cost of training large models from scratch.

MarianMT, developed by the Microsoft Translator team, is designed to provide an effective balance between translation quality and computational efficiency, which is well-suited for research contexts with limited resources (Junczys Dowmunt et al., 2018). By developing a bilingual translation system

explicitly tailored to Ilocano and Tagalog, this study contributes to ongoing efforts to expand linguistic inclusion in digital technologies. More broadly, it aims to support Ilocano-speaking communities in accessing digital information and services while providing a methodological foundation for future work on other underrepresented Philippine languages.

Objectives of the Study

This study aims to develop a neural machine translation model for Ilocano-Tagalog translation by fine-tuning the MarianMT architecture in a low-resource language setting, and to analyze model behavior using controlled experimental evaluation. Specifically, the study seeks to:

1. Gather and preprocess an Ilocano-Tagalog parallel corpus, including systematic data collection, cleaning, normalization, and controlled mask-based data augmentation.
2. Fine-tune MarianMT-based translation models using augmented and non-augmented datasets to examine the effect of data augmentation on model learning.
3. Assess translation output quality using automatic metrics, including BLEU, ChrF, and METEOR, under both human reference and dataset target reference evaluation conditions.
4. Evaluate model behavior through controlled experimental analysis, including ablation studies, sentence-length-based performance analysis, comparison with benchmark translation systems, human-assisted evaluation using fluency, adequacy, and faithfulness criteria, and statistical analysis of sentence-level performance differences.

Significance of the Study

This study examines the potential of fine-tuning the Marian Machine Translation architecture for Ilocano and Tagalog neural machine translation in a low-resource language setting. Ilocano and Tagalog are two major Philippine languages that remain underrepresented in digital and computational resources, particularly in the availability of high-quality parallel corpora and language technologies. By developing and evaluating a baseline Ilocano and Tagalog neural machine translation system, this research helps address existing linguistic gaps in natural language processing and digital communication involving Philippine languages.

From a technical and research perspective, the study provides empirical evidence on how data augmentation strategies and training configurations influence model behavior in low-resource neural machine translation. Through controlled experimental evaluation, including ablation studies, sentence-length-based comparisons, benchmark-system comparisons, and statistical analyses, the study offers methodological insights into neural machine translation performance under varying evaluation conditions. The inclusion of both human reference and dataset target reference evaluations demonstrates the limitations of relying solely on automatic evaluation metrics and highlights the importance of careful evaluation design for low-resource language pairs.

The findings of this study align with global and national initiatives that promote linguistic diversity and inclusion in digital technologies. In particular, the research supports UNESCO’s advocacy for indigenous and minority languages in information technologies and contributes to the objectives of the

International Decade of Indigenous Languages from 2022 to 2032, which emphasizes language preservation as a component of sustainable development (UNESCO, 2022). The study is also relevant to the United Nations Sustainable Development Goals, specifically SDG 4 on quality education, SDG 10 on reduced inequalities, and SDG 16 on peace, justice, and strong institutions, by increasing the digital visibility and computational support available for Ilocano.

In the education sector, regional education offices such as the Department of Education regional and division offices in the Ilocos Region and other provinces with Ilocano-speaking populations, as well as Ilocano-medium public and private schools, may benefit from the outcomes of this research. The translation model developed in this study may serve as a prototype tool to support the preparation and translation of instructional materials, school communications, and learning resources between Ilocano and Tagalog, subject to human review. These tools can assist educators in developing linguistically accessible and culturally appropriate learning materials while maintaining instructional accuracy, in line with the goals of the Mother Tongue Based Multilingual Education policy.

Local Government Units, Public Information Offices, and government outreach agencies may also benefit from advances in neural machine translation for Ilocano and Tagalog. Public institutions such as the Department of Social Welfare and Development and the Department of Health regularly disseminate information related to health services, emergency preparedness, education, and social programs. The translation model developed in this study may serve as a supporting tool in document preparation and communication

workflows to assist with translating public advisories and informational materials, with human validation to ensure accuracy and contextual appropriateness. Improved access to multilingual communication tools can support clearer information dissemination, enhance transparency, and promote inclusive public engagement.

For future academic researchers and practitioners in natural language processing, artificial intelligence, linguistics, and digital humanities, this study provides a reproducible case study and methodological reference for working with low-resource language pairs. The evaluation framework, data preparation procedures, and comparative analysis presented in this research may inform future studies and system development efforts involving other underrepresented Philippine languages such as Pangasinense, Kapampangan, and Tausug. By demonstrating a careful and evaluation-driven approach to neural machine translation development, this study contributes to ongoing efforts toward ethically responsible, culturally informed, and linguistically inclusive language technologies.

Scope and Limitation of the Study

This study presents and experimentally evaluates a text-based Ilocano-Tagalog neural machine translation system based on the Marian Machine Translation (MarianMT) architecture. The primary objective is to develop and evaluate a benchmark translation model for this low-resource Philippine language pair using a parallel Ilocano-Tagalog corpus. By leveraging MarianMT's computational efficiency and suitability for language-specific fine-tuning, the study establishes a foundational translation model that may support

future research and resource development for other regional Philippine languages.

The scope of the study is limited to unidirectional translation from Ilocano to Tagalog. This design choice reflects the low-resource nature of Ilocano, which is characterized by limited availability of parallel corpora, linguistic tools, and digitized text resources. Prioritizing the Ilocano-to-Tagalog direction aligns with the study's goal of improving linguistic accessibility by enabling Ilocano speakers to access information in a widely used national language. Focusing on a single translation direction also allows for more controlled fine-tuning, systematic experimentation, and detailed analysis of model behavior without compromising the validity of the findings. The reverse translation direction (Tagalog to Ilocano) is outside the scope of this study and is left for future work when sufficient data becomes available.

This research is experimental in nature and does not aim to deploy a production-ready translation system. Translation outputs are assessed using automatic evaluation metrics, including BLEU, ChrF, and METEOR, computed under both human-reference and dataset-target-reference conditions. These metrics are used solely for assessing translation output. Model behavior is analyzed through controlled experimental evaluation, including ablation analysis between augmented and non-augmented training data, sentence-length-based comparisons, benchmark system comparisons, and statistical analysis of sentence-level performance differences. Human-assisted evaluation using fluency, adequacy, faithfulness, and overall translation quality criteria is included as qualitative validation to support the interpretation of automatic metric results.

The study employs a single MarianMT-based architecture and does not explore alternative model architectures, large-scale multilingual systems, or external knowledge integration techniques. Data augmentation is limited to mask-based augmentation, and no additional cross-lingual transfer learning methods or external linguistic resources are incorporated beyond those inherent to the pre-trained MarianMT model. This design reflects a deliberate focus on controlled experimentation rather than architectural optimization.

Several limitations are acknowledged. First, the size of the parallel corpus is relatively small, reflecting typical constraints in low-resource language research. Second, domain coverage is limited, as the datasets consist primarily of conversational and general-domain text, including informal exchanges and publicly available online content. As a result, the model's performance may not generalize well to specialized or formal domains, such as legal, medical, scientific, or technical texts, which require domain-specific terminology and discourse structures. Third, the evaluation focuses exclusively on text-based translation and does not consider speech translation, multimodal inputs, or real-time translation scenarios.

While MarianMT offers practical advantages in terms of computational efficiency and accessibility, the model is comparatively lighter than larger multilingual models. Consequently, the findings of this study are constrained by both data availability and model capacity. Despite these limitations, the scope reflects realistic conditions for low-resource neural machine translation research. It provides meaningful insights into the feasibility of fine-tuning MarianMT for Ilocano-Tagalog translation and for designing evaluation frameworks appropriate to low-resource language settings.

CHAPTER 2

THEORETICAL FRAMEWORK OF THE STUDY

Review of Related Literature and Studies

This chapter discusses pertinent literature and studies that serve as theoretical and empirical foundations for the research. Previous studies in neural machine translation, data augmentation, and evaluation are reviewed, with particular emphasis on low-resource languages such as Ilocano and Tagalog. Local and international studies are examined to identify existing gaps and limitations in current research, which inform the development of an improved Ilocano-Tagalog translation model using MarianMT. Overall, this review situates the study within the broader field of multilingual translation research and clarifies the objective of improving translation quality through enhanced fine-tuning and systematic evaluation.

Philippine Multilingualism and the Relevance to NLP and Machine Translation

The Philippines is one of the most linguistically diverse countries in the world, with more than 180 languages spoken across its regions (Eberhard, Simons, & Fennig, 2020; SIL International, n.d.). Despite this diversity, digital language technologies have historically prioritized a small number of dominant languages, particularly English and Tagalog, resulting in uneven access to language tools and resources for regional languages (Liu et al., 2020; Tiedemann, 2020). This imbalance has implications beyond technology, as language access influences participation in online discourse, access to information, and engagement with public services (Roche, 2023). Studies note

that communities whose languages are underrepresented in NLP systems often face barriers in education, governance, and digital communication, reinforcing existing social and regional inequalities (Joshi et al., 2020). These conditions motivate research on machine translation for Philippine languages, particularly in low-resource settings.

Defining Low-Resource Languages in the Philippine Context

Low-resource language translation refers to machine translation for language pairs with limited parallel corpora, scarce annotated datasets, and insufficient computational resources, regardless of the number of speakers (Joshi et al., 2020). In the Philippine context, this definition applies to many widely spoken languages, including Tagalog and Ilocano, because resource scarcity is driven by data availability rather than population size (Miranda, 2022; Pakray et al., 2025). Cruz and Cheng (2021) emphasize that regional Philippine languages such as Ilocano, Waray, and Cebuano remain underserved in MT research, with limited benchmarking and evaluation efforts. As a result, controlled experimentation and careful evaluation design are necessary to establish baseline translation performance and to examine the effects of training strategies in these settings.

Linguistic Characteristics of Tagalog and Ilocano Relevant to MT

Tagalog belongs to the Central Philippine or Mesophilippine subgroup and is closely related to Bicol and Bisayan languages such as Cebuano and Hiligaynon (Encyclopedia Britannica, 2025). Filipino is derived from Tagalog and incorporates lexical elements from other Philippine languages as well as Spanish and English, leading to variation that affects normalization,

tokenization, and translation consistency (Ethnolink, 2021). Although Tagalog has greater digital visibility than many regional languages, it remains computationally low-resource due to limited high-quality annotated corpora and structured NLP tools compared with major global languages (Miranda, 2022).

Ilocano is commonly classified as a member of the Cordilleran subgroup of Philippine languages and is spoken by millions of Filipinos, primarily in northern Luzon (Encyclopedia Britannica, 2025). Despite the widespread use, Ilocano remains underrepresented in NLP research and has fewer computational resources available for data-driven approaches (Casil, 2018; Pakray et al., 2025). Linguistic features such as stress-based meaning distinctions, glottal stops, and productive reduplication contribute to morphological and phonological complexity that can influence machine translation behavior (Boodon, 2022). These characteristics make Ilocano-Tagalog translation a suitable case for studying low-resource NMT adaptation.

Translation Studies Perspectives and Implications for MT Evaluation

Translation research highlights that meaning is shaped by linguistic form as well as contextual and cultural factors, and mismatches between these factors can lead to interpretation errors (Hatim & Munday, 2019). Janković-Paus (2017) explains that messages are produced within culturally specific communicative norms, which may not transfer directly across languages, increasing the risk of misunderstanding. Cultural expressions, idioms, and figurative language present additional challenges, as they often resist direct translation and require adaptive strategies that may alter tone or emphasis (Bassnett, 2014).

Translation output is also inherently variable. Pym (2015) notes that translator choices are influenced by style, audience, and communicative goals, resulting in multiple acceptable translations for a single source text. This variability complicates automatic evaluation, which often assumes a single reference translation. Ethical considerations further affect translation practice, as translators and translation systems can shape meaning through omission, emphasis, or framing, particularly in sensitive domains (Bielsa, 2020). These issues support the use of both automatic metrics and human evaluation in machine translation research.

Multilingual Pretraining and Fine-Tuning for Low-Resource MT

Large multilingual translation systems often rely on extensive training data and substantial computational resources, which limits their effectiveness for low-resource language pairs with narrow or uneven domain coverage (Ranathunga et al., 2021). Fine-tuning pretrained models on language-specific data is, therefore, a common strategy for low-resource MT. MarianMT, developed by the Helsinki NLP group, provides pretrained bilingual models trained on OPUS corpora and supports efficient fine-tuning using the Marian C++ toolkit (Junczys-Dowmunt et al., 2018).

In parallel, large-scale initiatives such as Meta AI's No Language Left Behind project demonstrate a broader research trend toward inclusive multilingual translation by supporting hundreds of languages in a single model (NLLB Team, 2022). Evaluation resources such as FLORES-101 provide standardized benchmarks that facilitate comparison across low-resource language settings (Goyal et al., 2022). These developments establish the

foundation for targeted Ilocano-Tagalog translation research using controlled fine-tuning and evaluation.

Training Strategies for Low-Resource Settings

Transformer-based architectures enable improved modeling of long-range dependencies and form the basis of most contemporary NMT systems (Bahdanau et al., 2014; Vaswani et al., 2017). In multilingual contexts, fine-tuning and transfer learning enable models to leverage knowledge from higher-resource languages (Liu et al., 2020). Parameter-efficient approaches, such as adapter-based fine-tuning, aim to reduce computational cost while preserving pretrained representations and have shown positive effects in multilingual and low-resource scenarios (Chronopoulou et al., 2023; Sel & Hanbay, 2024; Lankford, Afli, & Way, 2023). Domain adaptation is particularly relevant for low-resource corpora, which are often limited to conversational or general-domain text. Studies demonstrate that pretrained models can adapt to new domains using limited annotated data through specialized fine-tuning strategies (Zhang & Zong, 2020; Liang et al., 2021). Data augmentation further addresses data scarcity by increasing linguistic variability. Mask-based augmentation generates alternative sentence pairs by masking tokens or spans and predicting replacements using masked language models, producing semantically similar variants without collecting new corpora (Kumar, Choudhary, & Cho, 2020). Controlled masking strategies have shown measurable improvements in translation quality under low-resource conditions (Cheng, Huang, & Duan, 2022).

Evaluation Design for Ilocano-Tagalog MT

Automatic evaluation metrics enable efficient comparison of MT systems but require careful interpretation in low-resource and morphologically rich settings. BLEU measures n-gram overlap between system output and reference translations and remains widely used due to efficiency and comparability (Papineni et al., 2002), but shows limited sensitivity to paraphrasing and morphological variation and weaker correlation with human judgment at the sentence level, particularly in low-resource contexts (Post, 2018; Reiter, 2018). METEOR combines precision, recall, and flexible matching strategies and often correlates more closely with human judgments, though reliance on language-specific resources may limit applicability in under-resourced settings (Banerjee & Lavie, 2005; Lavie & Denkowski, 2009; Callison-Burch et al., 2006; Denkowski & Lavie, 2014). ChrF evaluates character-level n-gram overlap and effectively captures morphological variation and partial word matches in languages such as Ilocano and Tagalog (Popović, 2015; Popović, 2016; Sennrich et al., 2016). Human evaluation complements automatic metrics by assessing adequacy, faithfulness, and fluency beyond automatic scores (Snover et al., 2006; Castilho et al., 2018). Although human evaluation involves subjectivity and resource constraints, inter-annotator agreement measures improve reliability, and Likert-scale designs support consistent scoring and quantitative analysis of subjective judgments (Escribe, 2019; Bojar et al., 2016; Boone & Boone, 2012; Joshi et al., 2015; Moorkens et al., 2018). Sentence-length-based analysis further supports evaluation by examining model behavior across varying levels of linguistic complexity (Bahdanau, Cho, & Bengio, 2015; Koehn & Knowles, 2017; Popović, 2018).

Literature Matrix

Table 1. Literature Matrix

Author(s), Year	Focus / Title	Methodology	Key Findings / Relevance to the Study
Pakray et al. (2025)	Low-resource Asian MT	Comparative analysis	Demonstrated that many Asian languages remained computationally low-resource due to limited parallel data, reinforcing the need for controlled experimentation and careful evaluation
NLLB Team (2022)	Multilingual MT	Large-scale multilingual training	Demonstrated a global trend toward inclusive machine translation for low-resource languages
Goyal et al. (2022)	FLORES-101 benchmark	Dataset development	Provided standardized evaluation benchmarks for low-resource machine translation
Cheng, Huang & Duan (2022)	Controlled masking augmentation	Conditional MLM	Found that moderate masking improved translation quality in low-resource machine translation
Miranda (2022)	Philippine NLP resources	Review study	Identified limited annotated corpora and computational tools for regional Philippine languages
Cruz & Cheng (2021)	Philippine languages in MT	Empirical review	Showed that Ilocano and related languages remained underrepresented in machine translation benchmarking and evaluation
Eberhard, Simons & Fennig (2020)	Philippine language diversity	Descriptive survey	Documented linguistic diversity in the Philippines, motivating NLP research for underrepresented languages
Joshi et al. (2020)	Low-resource language definition	Analytical survey	Defined low-resource status by data scarcity rather than the number of speakers, supporting the classification of Ilocano and Tagalog as low-resource
Liu et al. (2020)	Multilingual transfer learning	Empirical study	Demonstrated that fine-tuning pretrained models benefited low-resource languages
Zhang & Zong (2020)	Domain adaptation	Review study	Highlighted the importance of adapting pretrained models to limited-domain corpora

Kumar, Choudhary & Cho (2020)	Data augmentation for MT	Experimental study	Demonstrated improved machine translation performance using mask-based augmentation
Castilho et al. (2018)	Human MT evaluation	Mixed-method evaluation	Identified adequacy, fluency, and faithfulness as core dimensions of translation quality
Junczys-Dowmunt et al. (2018)	MarianMT framework	Framework design	Introduced MarianMT as an efficient and trainable machine translation system suitable for fine-tuning
Koehn & Knowles (2017)	Sentence length effects	Empirical analysis	Showed that translation difficulty increased with sentence length
Vaswani et al. (2017)	Transformer architecture	Model design	Established transformer-based architectures used in modern neural machine translation systems
Popović (2015, 2016)	ChrF metric	Character-level evaluation	Demonstrated that ChrF was effective for morphologically rich languages such as Ilocano and Tagalog
Bahdanau et al. (2014)	Neural machine translation	Neural architecture	Introduced attention-based neural machine translation, forming the foundation of transformer models
Boone & Boone (2012)	Likert-scale analysis	Quantitative survey	Supported the use of Likert scales for structured linguistic evaluation
Snover et al. (2006)	Human vs. automatic evaluation	Empirical study	Demonstrated the need to complement automatic evaluation metrics with human judgment
Banerjee & Lavie (2005)	METEOR metric	Metric design	Introduced METEOR and showed improved correlation with human judgments compared to BLEU
Koehn (2004)	Statistical significance testing	Bootstrap resampling	Established paired bootstrap resampling for machine translation system comparison
Papineni et al. (2002)	BLEU metric	Evaluation metric	Introduced BLEU for automatic machine translation evaluation and established it as a widely adopted standard

Synthesis and Relevance of the Study

This chapter provides a systematic integration of theoretical and empirical foundations supporting the development of an Ilocano-Tagalog neural machine translation (NMT) system. Reviewed literature links linguistic context, model architectures, fine-tuning strategies, and evaluation practices to the challenges and opportunities in low-resource translation, with an emphasis on contemporary neural approaches, including MarianMT (Junczys-Dowmunt et al., 2018).

The Philippine linguistic landscape exhibits extensive linguistic diversity, often within the Austronesian language family. Tagalog and Ilocano function as widely used Philippine languages; however, limited digital corpora, linguistic tools, and annotated datasets constrain progress in data-driven translation research for both languages (Miranda, 2022; Pakray et al., 2025). Limited resource availability motivates methods focused on efficient learning under constrained data conditions (Joshi et al., 2020).

Machine translation research evolves from rule-based and statistical approaches toward neural systems, with transformer-based models enabling strong sequence modeling through attention mechanisms and parallel computation (Bahdanau et al., 2014; Vaswani et al., 2017). Transformer architectures underpin multilingual translation systems such as mBART and M2M-100, supporting transfer learning and multilingual fine-tuning for under-resourced language settings (Liu et al., 2020). Multilingual neural machine translation further supports cross-lingual transfer by leveraging high-resource language signals to improve performance on low-resource languages.

Fine-tuning, parameter-efficient adaptation, and domain-aware training remain central strategies for improving translation quality. Fine-tuning updates a pretrained model toward targeted linguistic or domain-specific objectives, whereas domain adaptation improves contextual relevance under shifting subject matter (Zhang & Zong, 2020; Liang et al., 2021). MarianMT supports targeted bilingual fine-tuning through modular bilingual checkpoints trained on OPUS resources (Junczys-Dowmunt et al., 2018). Mask-based augmentation further addresses data scarcity by generating semantically consistent sentence variants and expanding training diversity under limited parallel data conditions (Kumar et al., 2020; Cheng et al., 2022; Li et al., 2022).

Model reliability in low-resource settings requires attention to overfitting and generalization, supported by regularization, dropout, and early stopping techniques (Bejani & Ghatee, 2021; Balestriero et al., 2022; Piotrowski & Napiorkowski, 2012). Translation quality assessment benefits from the combined use of automatic metrics and human evaluation dimensions such as adequacy, fluency, and faithfulness (Snover et al., 2006; Castilho et al., 2018; Läubli et al., 2020). Likert-scale scoring supports conversion of subjective judgments into analyzable quantitative ratings (Boone & Boone, 2012; Joshi et al., 2015; Moorkens et al., 2018). Statistical validation of automatic metric differences is supported through bootstrap sampling for machine translation evaluation under non-parametric conditions (Koehn, 2020; Freitag et al., 2021).

Community-oriented annotation and language-resource development further strengthen culturally sensitive and context-appropriate translation research. Participatory approaches, as demonstrated in initiatives such as

Masakhane, highlight the value of native-speaker collaboration for data validation and linguistic authenticity (Nekoto et al., 2020).

Conceptual Framework

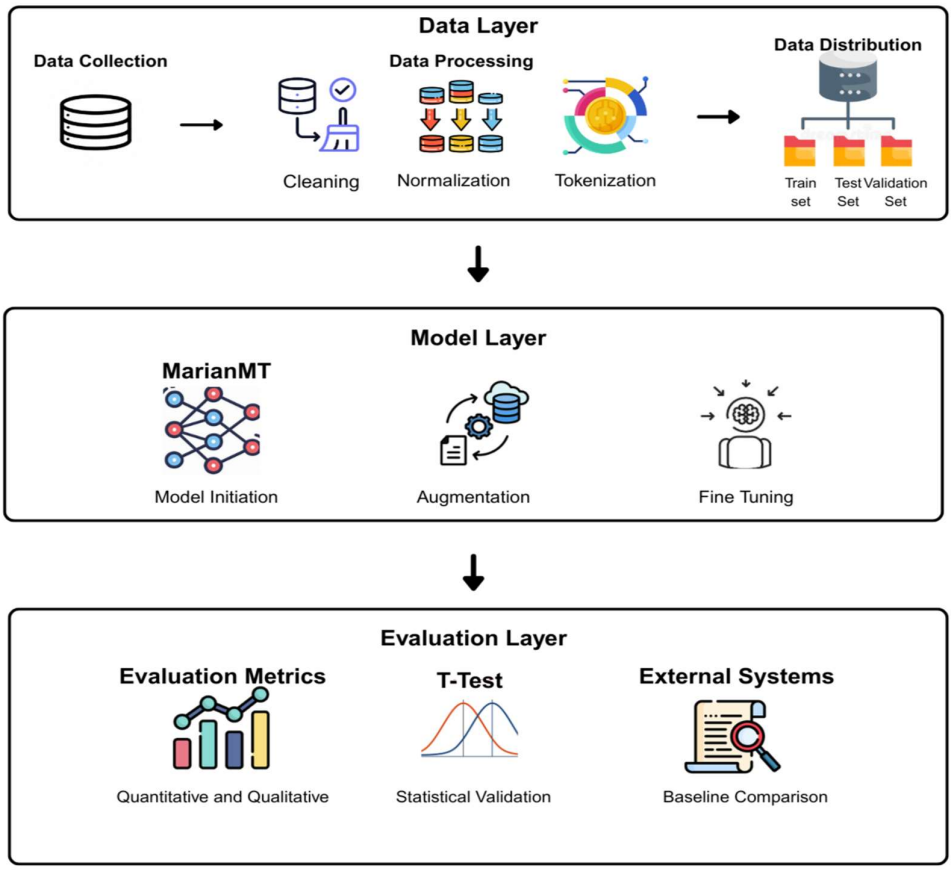


Figure 1. Conceptual Framework of the Study

This study develops a neural machine translation (NMT) system specifically for the Ilocano-Tagalog language pair, which remains underrepresented in natural language processing research. As shown in Figure 1, the conceptual framework comprises three phases: data preparation, model training, and evaluation and analysis. Each stage supports the goal of examining translation outputs and analyzing model behavior under controlled experimental conditions in a low-resource setting for the Philippine language.

In the initial stage of the study, an Ilocano-Tagalog parallel corpus was compiled, consisting of approximately 42,000 sentence pairs. These sentence pairs were collected from multiple sources, including researcher-authored sentences, conversational and general-domain text, and publicly available online materials. Additional parallel texts were obtained from online archival sources to supplement the corpus. Following data collection, quality assurance procedures were conducted, which involved manual inspection, cleaning, and correction of misaligned or problematic sentence pairs. After these steps, approximately 10,000 high-quality sentence pairs were retained for model training and evaluation. The dataset was then normalized and tokenized using SentencePiece to segment text into subword units, enabling more efficient vocabulary representation, particularly in a low-resource setting. The finalized dataset was first partitioned into training, validation, and test sets to ensure that evaluation data remained independent. Mask-based data augmentation was then applied only to the training set using a strategy inspired by masked language modeling to create additional training variants and increase data variability. The training set was used to update model parameters, the validation set was used for monitoring during training, and the test set was reserved for final evaluation.

In the second phase, model training was conducted using the MarianMT implementation of the OPUS-MT model, specifically Helsinki-NLP's opus-mt-en-tl checkpoint, which was further fine-tuned on the Ilocano-Tagalog parallel corpus. This fine-tuning process allows the model to adapt to the target language pair's syntax, vocabulary, and sentence patterns. Training was performed in the Google Colab environment using the Hugging Face

Transformers library. Model parameters such as learning rate, batch size, training steps, and evaluation intervals were carefully configured. Given the limited dataset size, training was closely monitored to reduce overfitting while enabling the model to learn meaningful linguistic patterns. Separate training runs were conducted using augmented and non-augmented training data to support controlled comparison.

The third phase involves evaluation and analysis of the translation outputs and model behavior. Translation output quality was assessed using automatic evaluation metrics, including BLEU, ChrF, and METEOR, and compared with outputs from external systems such as Google Translate and ChatGPT. Statistical analysis was performed using paired bootstrap resampling on sentence-level evaluation scores to estimate confidence intervals and assess whether observed differences between model variants were statistically meaningful (Koehn, 2004). In addition to quantitative metrics, the study includes human-assisted qualitative evaluation of selected translation outputs using Likert-scale ratings for adequacy, fluency, and faithfulness, providing interpretive support for the automatic evaluation results.

This conceptual framework guides the systematic development and experimental evaluation of an Ilocano-Tagalog neural machine translation system trained on limited but carefully prepared data. By integrating structured data preparation, focused model fine-tuning, and layered evaluation procedures, the framework provides insights into how neural machine translation models behave in low-resource settings. It contributes to ongoing efforts to develop language technologies for underrepresented Philippine languages.

CHAPTER 3

METHODOLOGY

Study Area

The research was carried out at strategic sites selected based on accessibility, available resources, and proximity to the research requirements. The Laguna State Polytechnic University (LSPU) Los Baños Campus in Laguna was selected due to its well-stocked library, which contains a vast collection of books, theses, and journals necessary for reviewing related literature. The College of Computer Studies (CCS) Faculty Offices at LSPU-LB were used for expert consultations, during which the researchers tested methods and improved the academic quality of the study.

The LSPU-Los Baños computer laboratory and student lounges were also used for online database access, document preparation, and collaborative discussions. In addition, online academic platforms such as Google Scholar, ResearchGate, and Academia.edu were extensively used to obtain up-to-date scholarly content. These platforms provided access to relevant academic materials, expert insights, and supporting resources that contributed to the successful completion of the research.

Materials

The resources utilized in this research encompass both hardware and software components necessary for the development, training, and evaluation of the proposed neural machine translation model. These materials support all

stages of the research workflow, including data preparation, model implementation, experimentation, and performance assessment.

Software

The software resources employed in this study were the necessary tools, frameworks, and platforms for implementing, training, and evaluating the machine translation model. This included programming environments, machine learning libraries, and evaluation toolkits that supported the implementation of the model, as well as data preprocessing, fine-tuning, and performance evaluation.

Table 2. Software Specification

Components	Version and Details
Operating System	Ubuntu 20.04 LTS (Google Colab runtime)
Programming Language	Python 3.10
Deep Learning Framework	PyTorch 2.x
NLP Framework	Hugging Face Transformers 4.x
Tokenizer	SentencePiece
GPU Acceleration	CUDA 11.x (Colab)

Python 3.10

Python 3.10 served as the primary programming language for this study, providing a stable and flexible environment for implementing the complete machine translation pipeline. The modern syntax and performance improvements facilitated clear, maintainable code across all stages of the research, including data preprocessing, corpus management, model fine-tuning, and evaluation. In addition, Python 3.10 offers extensive compatibility with widely used natural language processing and deep learning libraries, enabling

seamless integration of tools for tokenization, model training, and automatic evaluation.

Google Colab

Google Colab was used as the primary development and execution platform for this study, providing a cloud-based environment suitable for training and evaluating neural machine translation models. The built-in access to GPU resources enabled efficient fine-tuning and testing of the model without requiring high-end local hardware, making large-scale experimentation feasible under resource constraints. Additionally, Google Colab supported collaborative coding, easy integration with Python-based machine learning libraries, and reproducible experiment execution through shared notebooks and controlled runtime configurations. This setup enabled the researchers to iteratively refine models and conduct evaluations in a consistent, accessible computing environment.

MarianMT (Hugging Face Transformers)

MarianMT served as the base neural machine translation model fine-tuned for Ilocano-Tagalog translation. MarianMT is a pre-trained Transformer-based model designed for low-resource language translation. In this study, the model was fine-tuned using a parallel Ilocano-Tagalog corpus to improve translation quality. Model performance was evaluated using automatic metrics, including BLEU, ChrF, and METEOR, as well as human-assisted evaluation. Model training and inference were managed through the Hugging Face Trainer utilities.

Libraries

The following libraries were utilized to support the implementation, fine-tuning, evaluation, and analysis phases of the study. These libraries provided essential functionality for data preprocessing, training neural machine translation models, automatic evaluation using standard metrics, and result analysis. Their selection was based on reliability, widespread adoption in natural language processing research, and compatibility with the experimental framework employed in this study.

SentencePiece

SentencePiece was employed for subword tokenization to ensure consistent segmentation across the parallel corpus. This step was crucial in the low-resource setting, where maximizing vocabulary coverage from limited data is critical.

Hugging Face Transformers

The Hugging Face Transformers library was used to load the Helsinki-NLP/opus-mt-en-tl MarianMT model and control the fine-tuning process. This provides tools for training routines, evaluation utilities, and seamless integration with the Seq2SeqTrainer API for low-resource translation tasks.

Hugging Face Datasets

The Hugging Face Datasets library was utilized to load, structure, and split the Ilocano-Tagalog corpus into training, validation, and test sets. The

integration with the Transformers library enabled efficient preprocessing, dataset handling, and reproducible experimentation.

Evaluate with SacreBLEU

The Hugging Face Evaluate library, in conjunction with SacreBLEU, was used to compute BLEU and ChrF scores. This combination provided a standardized and reproducible framework for comparing machine-generated translations with human reference translations.

NLTK (METEOR)

NLTK was used to compute the METEOR score for assessing translation output. METEOR complemented BLEU and ChrF by accounting for unigram matches, morphological variation, and paraphrastic similarity, offering an additional perspective on translation quality.

PyTorch

PyTorch served as the core deep learning framework, responsible for constructing computation graphs, performing backpropagation, and handling optimization, enabling efficient batching operations and GPU-accelerated training during model fine-tuning.

NumPy

NumPy was utilized for numerical computations throughout the study, particularly for efficient array manipulation and mathematical operations. This played a key role in processing model outputs, reference translations, and intermediate data structures during evaluation. In addition, NumPy supported

the implementation of custom metric calculations and aggregation of evaluation results.

Pandas

Pandas was employed to load, manage, and preprocess datasets used in the study. It was beneficial for handling CSV and Excel files during data cleaning, filtering, and augmentation procedures. The library also enabled efficient inspection, transformation, and organization of parallel sentence pairs prior to model training and evaluation.

TensorBoard

TensorBoard was used to track and visualize the fine-tuning process. Metrics such as training loss, validation loss, and BLEU scores were logged to monitor model convergence and performance trends across epochs. Visualizing these metrics helped identify overfitting or underfitting early, allowing timely adjustments to hyperparameters.

Matplotlib

Matplotlib was utilized to generate plots for training curves and evaluation results, providing explicit visual representations of model behavior and performance.

Hardware

The study was conducted using personal computing equipment capable of handling natural language processing tasks, including model fine-tuning, inference, and evaluation. To access additional computational resources when

required, cloud-based hardware via Google Colab was used, providing access to high-performance GPUs for efficient training.

Table 3. Desktop Specification

CPU	Intel Celeron N4100 CPU @ 1.10GHz
RAM	8.0 Gigabytes
GPU	Intel UHD Graphics 600
STORAGE	256 Gigabytes HDD/SSD

Research Design

This study employed a quantitative experimental research design to evaluate the effects of mask-based data augmentation on the MarianMT model for Ilocano-Tagalog translation. The experiment was conducted in a digitally controlled environment, with training conditions systematically manipulated to examine differences in translation output quality. The primary objective was to determine whether a fine-tuned MarianMT model trained with mask-based data augmentation exhibited measurable performance differences compared to a non-augmented baseline model and selected existing translation systems.

The independent variable in this study was the application of mask-based data augmentation during model training. In contrast, the dependent variables comprised translation quality scores obtained from automatic evaluation metrics, including BLEU, ChrF, and METEOR, as well as results from human-assisted evaluation. Two experimental configurations were implemented: a baseline MarianMT model trained without augmentation and an augmented MarianMT model trained using mask-based data augmentation. Both models were trained under identical architectures, hyperparameters, and

optimization settings to ensure experimental consistency and isolate the effect of augmentation.

Quantitative evaluation was performed by assessing model outputs on a strictly held-out test set using BLEU, ChrF, and METEOR. Paired bootstrap resampling was applied to estimate the reliability of observed performance differences between the augmented and non-augmented models. To complement automatic evaluation, subset-based analyses were conducted using human-translated references. These included automatic metric computation against human translations, human-assisted evaluation using Likert-scale ratings of adequacy, fluency, and faithfulness, and manual error annotation to identify common translation error patterns. These subset-based evaluations were treated as supplementary diagnostic analyses intended to support interpretive insight rather than statistically generalizable claims.

System Architecture

The Ilocano-tagalog neural machine translation system follows a linear workflow that defines the flow of data from corpus input to result analysis. The architecture is structured to clearly separate data preparation, model training, output generation, and evaluation to support controlled experimentation in a low-resource setting.

The workflow begins with the Parallel Corpus, which contains aligned Ilocano-Tagalog sentence pairs. This corpus is processed through a Data Processing component responsible for cleaning, normalization, and subword tokenization using the SentencePiece tokenizer associated with MarianMT.

The processed data are then partitioned into training, validation, and test sets, with data augmentation applied exclusively to the training split.

The Model Fine-Tuning component adapts a pre-trained MarianMT model to the Ilocano-Tagalog language pair using the prepared training data. After fine-tuning, the system generates translations for unseen inputs in the Model Output stage.

Generated outputs are passed to an independent Evaluation component, which assesses translation quality using both automatic and human-assisted methods. The results are then consolidated in the Result Analysis and Visualization stage, where performance comparisons and summaries are produced for interpretation.

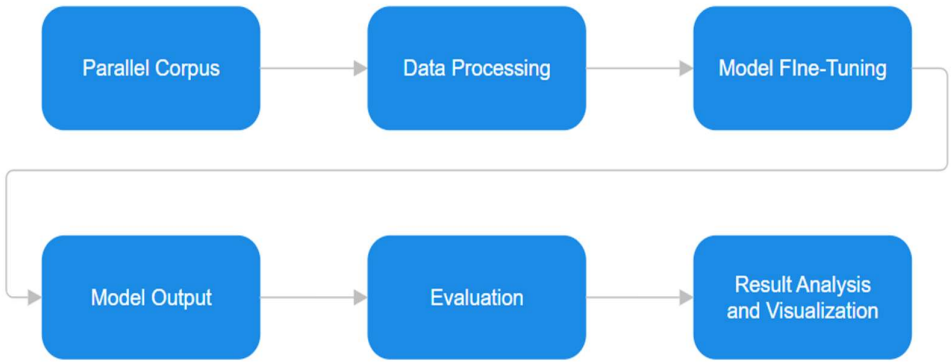


Figure 2. System Architecture

Model Development

Model development was conducted using MarianMT, which served as the core neural machine translation architecture in this study. MarianMT is based on the Transformer encoder-decoder model and is well-suited for low-

resource language experimentation due to its efficiency and support for multilingual fine-tuning.

Two MarianMT-based models were developed: a non-augmented baseline model, trained on 8,130 Ilocano-Tagalog sentence pairs, and an augmented model, trained on an expanded set of 32,520 sentence pairs generated through source-side mask-based data augmentation applied exclusively to the training data. Both models shared identical architectures, tokenization schemes, and training configurations to ensure fair comparison.

Model training involved tuning key hyperparameters, including learning rate, batch size, and the number of training steps. Training progress was monitored using validation loss and automatic evaluation scores to maintain stable learning behavior and reduce the risk of overfitting, given the limited size of the parallel corpus.

Following training, translation quality was evaluated using automatic metrics (BLEU, ChrF, and METEOR) computed on a held-out test set. To assess the statistical reliability of observed performance differences between model variants, bootstrap resampling was applied to sentence-level metric scores. This non-parametric method estimates confidence intervals by repeatedly sampling the test set with replacement and is appropriate for small datasets and non-normal score distributions.

In addition to a complete test-set evaluation, a subset-based analysis was conducted using 80 test sentences translated by human annotators. These human translations were treated as gold-standard references for computing automatic evaluation metrics and for comparing the outputs of the augmented

and non-augmented MarianMT models against other translation systems on the same inputs. A separate 195-sentence subset was used for human-assisted evaluation, in which model outputs were rated on a 5-point Likert scale for adequacy, fluency, and faithfulness, and for manual error annotation to identify and categorize common translation errors.

Overall, the model development process focused on controlled fine-tuning of MarianMT and systematic evaluation of data augmentation effects, supporting the study's objective of investigating neural machine translation for Ilocano-Tagalog under low-resource conditions.

Data Collection and Preprocessing

The dataset used in this study comprised approximately 42,000 Ilocano-Tagalog sentence pairs collected from both publicly available sources and researcher-developed corpora. Publicly sourced data included sentence-aligned religious texts from the Ilokano-to-Tagalog-Machine-Translation corpus (Cob, 2022) and bilingual text samples from online news websites, digital publications, and other internet-based sources. To increase coverage, the researchers also composed original sentences reflecting conversational and general-domain usage, including everyday interactions and contemporary public communication. Some sentence pairs contained mixed English and Tagalog elements, reflecting natural code-switching practices commonly observed in Philippine communication.

Following cleaning and filtering, which involved removing duplicate entries, noisy translations, and misaligned sentence pairs, the corpus was reduced to approximately 10,000 high-quality parallel sentence pairs.

Preprocessing steps included text normalization (punctuation, casing, and spacing) and subword tokenization using the pre-trained SentencePiece tokenizer within the MarianMT framework to ensure consistent representation and mitigate vocabulary sparsity.

The cleaned dataset was then divided into three subsets: 8,130 sentence pairs for training, 904 for validation, and 1,004 for testing. The dataset was split prior to augmentation to prevent data leakage. Mask-based data augmentation was applied only to the training set, wherein approximately 15% of SentencePiece subword units in the Ilocano source sentences were modified according to the data augmentation strategy described in the Data Augmentation Strategy subsection. This process expanded the training set from 8,130 to 32,520 sentence pairs, while validation and test sets remained unchanged. The resulting datasets were used to fine-tune and evaluate the MarianMT model.

Data Augmentation Strategy

To address data scarcity in Ilocano-Tagalog translation, this study employed mask-based data augmentation on the training split only. After the dataset was partitioned into training, validation, and test sets, the training dataset was augmented by generating masked variants of the Ilocano source sentences.

Masking was implemented at the word level. For each training example, the Ilocano sentence was split into whitespace-separated tokens, and each token had a 15% probability of being replaced with the placeholder token <mask>. Tokens not selected for masking were kept unchanged. The resulting

masked sentence was stored in a new input field (ilocano_masked) paired with the original Tagalog target sentence.

To increase variability, multiple masked variants were generated for each training sentence by repeating the masking process and concatenating the resulting datasets. Using three additional masking passes, the augmented training set contained approximately four times as many training examples as the original training split. Finally, empty or blank rows were removed from the augmented training set as well as from the validation and test sets to ensure that all examples used for training and evaluation contained non-empty source and target text.

Validation Setup and Data Splitting

The cleaned Ilocano-Tagalog parallel corpus was divided into training, validation, and test sets using random splitting with a fixed seed to ensure reproducibility and unbiased evaluation. The split was performed prior to any data augmentation, and augmentation was applied exclusively to the training set. The validation and test sets remained unchanged throughout preprocessing, training, and evaluation.

The training set was used solely for model fine-tuning. In contrast, the validation set served as a held-out subset to monitor learning progress and guide experimental decisions, such as selecting augmentation settings and training checkpoints. The test set was retained as an unseen holdout and was accessed only for final performance reporting and system-level comparison. This setup preserved strict separation between training and evaluation data and prevented information leakage across experimental stages.

Sentence-length grouping (short, medium, and long) was additionally applied during analysis to examine performance variation across different levels of input complexity. This supported a more straightforward interpretation of evaluation results by accounting for differences associated with sentence length.

Model Fine-Tuning

This study fine-tuned the Helsinki-NLP/opus-mt-en-tl MarianMT model using the Hugging Face Transformers framework in Google Colab to improve translation quality for the Ilocano-Tagalog language pair, and fine-tuning involved continuing the training of a pre-trained multilingual neural machine translation model on a smaller, task-specific parallel corpus, enabling the model to better adapt to the linguistic characteristics of the target language pair.

The Hugging Face training pipeline primarily managed the training process; however, key training parameters were carefully configured to balance performance, stability, and computational feasibility, given the limited dataset size and available hardware resources. These parameter choices allowed the model to learn meaningful translation patterns while reducing the risk of overfitting in a low-resource setting.

Model Configuration and Hyperparameter

Reproducibility and transparency of the experimental setup were ensured by explicitly specifying the configuration of the Marian Machine Translation (MarianMT) model used for Ilocano-Tagalog neural machine translation. The model was fine-tuned from a pre-trained MarianMT checkpoint

using supervised learning on a curated parallel corpus, with mask-based data augmentation applied exclusively to the training set.

Table 4. Model Configuration and Hyperparameter

Hyperparameter	Value
Model Architecture	MarianMT
Optimizer	AdamW
Learning Rate	5.00E-05
Epochs	5
Training Batch Size	32
Evaluation Batch Size	16
Dropout Rate	0.1
Loss Function	Cross-Entropy Loss

Learning Rate

The model used a learning rate of 5e-5, which controlled the magnitude of weight updates with each training step. If the learning rate was set too high, the model could become unstable and fail to converge; if set too low, training could proceed very slowly. The 5e-5 value was commonly used for fine-tuning transformer-based models because this provided a balance between training stability and convergence speed. During training, the Hugging Face training framework applied a learning rate scheduler that gradually reduced the step size as training progressed, allowing more refined parameter updates as the model approached convergence.

Batch Size

The training process was conducted with a batch size of 32, while a smaller size of 16 was used for evaluation. Batch size played a crucial role in determining training stability and memory usage. Larger batch sizes generally yielded more stable gradient updates but required greater GPU memory.

Due to the hardware limitations of the Google Colab environment, a maximum batch size of 32 was selected to balance gradient stability and memory availability, whereas evaluation employed a batch size of 16 to reduce memory load. Different batch sizes were used for training and evaluation to balance gradient stability during training and memory constraints during evaluation. This configuration enabled efficient optimization while ensuring that evaluation could be conducted within the hardware limitations of the Google Colab environment.

Epochs

The model was trained for five (5) epochs, meaning that the entire dataset was processed five times. An insufficient number of epochs could prevent the model from adequately capturing language-specific patterns. In contrast, excessive epochs increased the risk of overfitting, in which the model memorized the training data rather than generalizing to unseen inputs. Model performance was monitored at the end of each epoch, and checkpoints were saved to retain the best-performing model.

Beam Search (Decoding Strategy)

During translation generation, a beam search decoding strategy with a beam width of 5 was employed. Beam search allowed the model to explore multiple candidate translations simultaneously and select the sequence with the highest overall likelihood. By maintaining five alternative decoding paths at each step, the model achieved a balance between translation fluency and computational efficiency. This approach generally produced more fluent and

natural-sounding translations than greedy decoding, which selected only the most probable token at each step.

Weight Decay (Regularization)

A weight decay value of 0.01 was applied during training. Weight decay introduced a minor penalty to substantial weight values, thereby discouraging overly complex models and reducing the risk of overfitting. Given the limited dataset size, applying weight decay was particularly important, as the risk of overfitting was elevated in low-resource settings.

Optimizer

Model fine-tuning was performed using the AdamW optimizer, which was the default optimizer in the Hugging Face Trainer framework. AdamW was a variant of the Adam optimizer that decoupled weight decay from gradient-based learning-rate updates, thereby improving regularization. AdamW was widely used for training transformer-based models and was well-suited for multilingual and low-resource translation tasks. The ability to maintain stable convergence even with small batch sizes made this particularly effective for low-resource datasets.

Tokenizer Settings

The MarianMT model employed a pre-trained SentencePiece tokenizer, which segmented text into subword units rather than whole words. This approach was convenient for morphologically rich languages such as Tagalog and Ilocano, where words often contain complex affixes, reduplication, and inflectional variations. By operating at the subword level, the tokenizer reduced

vocabulary sparsity and minimized the out-of-vocabulary problem commonly encountered in low-resource language settings.

Subword tokenization enabled the model to represent rare or unseen words by recombining known subword units, which allows it to generalize beyond the exact forms observed during training. This contributed to more flexible lexical representation and improved handling of morphologically complex word forms. As a result, the model was better equipped to capture linguistic patterns across varying sentence structures, supporting more consistent translation performance across both common and less frequent lexical items.

Model Evaluation

The evaluation of the model referred to the different methods and measures of the proposed translation model's performance and accuracy, as is explained in the next section.

BLEU (Bilingual Evaluation Understudy)

Translation quality was evaluated using the BLEU metric, computed with the sacrebleu implementation in the Hugging Face evaluate library to ensure standardized and reproducible scoring. BLEU served as the primary automatic evaluation metric in this study because it provided an objective measure of translation quality based on n-gram overlap between machine-generated translations and reference translations.

BLEU quantified correspondence by computing modified n-gram precision across multiple n-gram lengths and applied a brevity penalty to

discourage overly short translations. Higher BLEU scores indicated greater lexical alignment with the reference translations.

During translation generation, the model employed beam search decoding with a beam width of five, enabling exploration of multiple candidate sequences before selecting the most probable output. This decoding strategy was chosen to improve translation fluency and coherence compared to greedy decoding, which selected only the most likely token at each step.

ChrF (Character n-gram F-score)

In addition to BLEU, the ChrF metric was used to evaluate translation outputs at the character level. ChrF computed an F-score based on character n-gram precision and recall, making the metric particularly suitable for morphologically rich languages such as Ilocano and Tagalog. Unlike word-based metrics,

ChrF was less sensitive to tokenization differences and better captured partial matches arising from affixation, inflection, and spelling variation. ChrF scores were computed using the sacrebleu implementation to ensure consistency and reproducibility.

METEOR (Metric for Evaluation of Translation with Explicit ORdering)

METEOR was employed as a complementary automatic evaluation metric to BLEU and ChrF. METEOR evaluated translation quality based on unigram alignment between candidate and reference translations, incorporating precision, recall, and a harmonic mean with greater emphasis on recall. The metric also accounted for word order and allowed flexible matching, which was

advantageous when evaluating translations that preserved meaning but differed in surface form. In this study, METEOR scores were computed using the NLTK meteor score implementation, providing an additional perspective on translation output similarity.

Statistical Significance Testing

To assess whether the observed differences between translation outputs were statistically meaningful, paired bootstrap resampling was applied to sentence-level BLEU scores following the methodology proposed by Koehn (2004). This analysis compared translation outputs produced by different systems on the same test sentences, ensuring that performance differences were evaluated under identical input conditions.

The bootstrap procedure repeatedly resampled the test set with replacement while preserving sentence-level alignment between system outputs. For each resampled set, BLEU scores were recomputed, and the difference between systems was recorded. This process was repeated 10,000 times to obtain an empirical distribution of BLEU score differences, from which confidence intervals and p-values were derived.

Paired bootstrap resampling did not assume normality of BLEU score distributions and was widely recommended for statistical comparison in machine translation evaluation, making this well-suited for the present study.

Human Evaluation

Human evaluation was conducted as a complementary assessment to automatic metrics, providing qualitative insight into translation quality. The

evaluation focused on adequacy, fluency, and faithfulness and was applied to a randomly sampled subset of 195 Ilocano-Tagalog sentence pairs from the held-out test set. These sentence pairs were translated using identical source inputs across both the augmented and non-augmented MarianMT model variants. Each translation was rated on a 5-point Likert scale (1 = Very Poor, 5 = Excellent) by two independent human evaluators, both native speakers of Ilocano and Tagalog, who had acquired both languages through early and sustained use in daily communication and professional contexts.

Evaluations were performed at the sentence level without access to model identity or reference translations, and no discussion or calibration was conducted during scoring to preserve evaluator independence. Mean scores for each dimension were computed by averaging ratings across the two evaluators. Adequacy measures the preservation of source meaning, fluency assesses grammatical correctness and naturalness in the target language, and faithfulness evaluates accurate conveyance of the source intent.

In addition to quality scoring, error annotation was performed on the same 195 translations to identify recurring issues, including mistranslation, omission, addition, grammatical errors, and lexical inconsistencies. Due to the detailed nature of this task, error annotation was conducted by a single evaluator to ensure consistency in error categorization.

Separately, a subset of 80 Ilocano sentences was used to construct a human reference set for cross-system comparison. These sentences were manually translated into Tagalog by one bilingual evaluator, and the results were used exclusively for automatic metric-based comparison of outputs from

the augmented model, non-augmented model, Google Translate, and ChatGPT. This subset was used solely for system-level comparison and does not replace the primary human evaluation conducted on the 195-sentence set.

Sample Size Determination for Human Evaluation

The number of sentence pairs selected for human evaluation was determined using a finite-population sampling approach on the held-out test set. This approach considered the total size of the test set and standard assumptions regarding variability and confidence commonly used in empirical evaluation.

Based on these considerations, a subset of 195 sentence pairs was selected to provide representative coverage of translation behavior while maintaining feasibility given the time and cognitive demands of sentence-level human assessment. The selected subset was drawn exclusively from the test set and was not used during training or validation, ensuring that human evaluation reflects model performance on unseen data.

Ethical Considerations

This study used only publicly available, non-sensitive textual data. All data sources were ethically obtained and properly credited throughout the research process. No personal, private, or personally identifiable information was collected or used in constructing the datasets. The study adhered to recognized ethical standards for data collection, processing, and reporting, emphasizing transparency, accountability, and reproducibility.

Human participation in the study was limited to evaluation and validation activities for assessing Ilocano-Tagalog machine translation outputs. Participation was conducted solely for academic research purposes. All contributors participated voluntarily and were informed of the objectives of the study, the nature of the evaluation tasks, and their right to withdraw at any time without penalty. Informed consent was obtained prior to participation.

For documentation and verification purposes, signed validation certification forms are included in the manuscript appendices. The inclusion of contributor names and signatures was obtained with the explicit consent of the participants and serves only to confirm the authenticity of the evaluation and validation process. These documents were not used for analytical purposes and were not linked to individual evaluation results or annotations.

All outputs obtained from human contributors, including evaluation scores, qualitative feedback, and reference translations, were treated as confidential. Data were anonymized prior to analysis and reported only in aggregated or de-identified form. Data handling, storage, and access control were conducted in accordance with the Data Privacy Act of 2012 (Republic Act No. 10173), with access restricted to the researchers to ensure participant privacy.

By documenting the methodology and evaluation process in detail, the study supports reproducibility and accountability in AI research. Ultimately, this work seeks to encourage the development of socially responsible, culturally sensitive, and beneficial language technologies for local communities. Furthermore, the study highlights the importance of safeguarding indigenous

and local languages in the digital space. This provides a foundation for future researchers to build ethically grounded and community-oriented AI systems.

The study acknowledges that machine translation systems may generate inaccurate or misleading translations, particularly in low-resource language settings. Potential risks include misinterpretation of meaning and loss of cultural nuance if outputs are used without verification.

To mitigate these risks, the Ilocano-Tagalog translation system was explicitly framed as a research prototype developed for experimental evaluation and not intended for public deployment or real-world application in the current form. Any practical use of machine-generated translations should involve appropriate human oversight.

Overall, this study aimed to promote responsible and ethical AI research by emphasizing ethical data use, voluntary participation, privacy protection, transparent evaluation procedures, and apparent limitations on system use, while advancing Philippine language technologies. This ensured that all datasets were collected, processed, and utilized in compliance with established ethical research standards and institutional guidelines.

The research also prioritized fairness and inclusivity by addressing biases commonly present in low-resource language datasets. Model performance and limitations were clearly reported to prevent misinterpretation or misuse of the system’s outputs.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter discusses the study's goals and the main findings and analysis. The researchers explored the fine-tuned MarianMT model for Ilocano-Tagalog translation, with an emphasis on the challenges of low-resource language processing, and included visual data and metrics to corroborate the findings.

Data Sources

The Ilocano-Tagalog corpus used in this study was constructed by aggregating sentence data from multiple publicly available repositories and researcher-generated resources. All sources were combined prior to any filtering, cleaning, or sentence-level alignment, to maximize coverage and diversity of linguistic expressions in both languages. The raw dataset was compiled from the following sources:

GitHub - JakeCob / Ilokano-to-Tagalog Machine Translation (CSV files)

This source consisted of Ilocano-Tagalog sentence data extracted from a student-developed machine translation project hosted on GitHub. This contributed 21,101 raw entries, primarily derived from instructional and experimental translation material.

GitHub - sheyn018 / SE-Ilokano-Tagalog-Translator repository

An additional 1,154 entries were obtained from a related Software Engineering language translator project. These entries included sentence

fragments and short translation units created for system testing and development.

OPUS - NLLB (Ilocano-Tagalog)

The OPUS No Language Left Behind (NLLB) corpus contributed 16,561 entries, mainly originating from religious and formal text domains. This source provided broad multilingual coverage but included long verse-style sequences and inconsistent segmentation.

Hugging Face - e-SALIN / Tagalog-to-Ilocano (mBART)

A smaller curated dataset of 457 entries was sourced from the e-SALIN project on Hugging Face, developed to support low-resource Philippine-language translation research**.**

Researcher-constructed sentence pairs

To supplement existing data and address domain gaps, the researchers manually constructed 3,000 sentence pairs. These sentences were designed to be grammatically correct, semantically aligned, and representative of everyday language use. This manual effort helped ensure coverage of expressions and vocabulary that are not well represented in publicly available corpora, contributing to a more balanced representation across different linguistic registers and domains.

In total, these sources yielded 42,273 raw sentence entries prior to cleaning and preprocessing. At this stage, the dataset included heterogeneous content with varying levels of alignment, sentence length, and linguistic quality. A detailed statistical description of the merged raw corpus was presented in the

following subsection, after which systematic cleaning and alignment procedures were applied.

Raw Data

The Ilocano-Tagalog corpus initially contained 42,273 entries from multiple sources, including some unpaired or duplicated sentences. Table 5 summarizes total words, characters, average words per sentence, and unique word counts. These statistics provide a baseline before cleaning, alignment, and filtering for model training.

Table 5. Descriptive Statistics of the Ilocano-Tagalog Parallel Corpus

Metric	Ilocano	Tagalog	Total
Sentence pairs	-	-	42,273 parallel pairs
Total words	752,031	862,621	1,614,652
Total characters	4,539,897	4,895,878	9,435,775
Average words per sentence	17.79	20.41	38.20 per sentence pair
Unique words	18,508	13,833	29,954

Table 5 summarizes the statistics of the combined Ilocano-Tagalog corpus prior to cleaning and alignment. At this stage, the dataset consisted of 42,273 raw entries collected from multiple sources and merged without enforcing strict sentence-level pairing. As a result, some Ilocano sentences appeared without corresponding Tagalog counterparts, which explains the higher lexical and character counts observed on the Ilocano side.

The raw corpus contained 752,031 Ilocano words and 862,621 Tagalog words, totaling 1,614,652 words and 9,435,775 characters across both languages. Differences in word and character distributions reflect both linguistic variation and the presence of unaligned entries prior to preprocessing.

Average word counts per sentence are reported separately for each language, with Ilocano averaging 17.79 words and Tagalog 20.41 words per entry. The combined average of 38.20 words per sentence represents the aggregate lexical length across source and target texts prior to alignment refinement. The raw corpus also exhibited substantial lexical diversity, containing 18,508 unique Ilocano words and 13,833 unique Tagalog words, for a total of 29,954 unique word types. This reflects the heterogeneous nature of the collected data and the absence of normalization, deduplication, and strict alignment at this stage.

These raw statistics provide a baseline description of the corpus before subsequent cleaning, sentence alignment, and filtering procedures, which enforce parallelism and produce the finalized dataset used for model training.

Data Cleaning and Preprocessing

Following inspection of the raw corpus, systematic data cleaning and preprocessing were applied to transform the merged entries into a sentence-level parallel dataset suitable for neural machine translation. At this stage, the corpus consisted of 42,273 raw entries aggregated from multiple sources.

Table 6. Corpus Preprocessing and Filtering Statistics

Stage	Before	After	Removed	Removed (%)
Remove empty/null / punctuation-only entries	42,273	39,994	2,279	5.39%
Remove duplicate sentence entries	39,994	39,070	924	2.31%
Remove extreme lengths (<2 or >13 words)	39,070	16,727	22,343	57.17%
Remove misaligned entries (length ratio > 1.33)	16,727	12,131	4,596	27.48%

Table 6 summarizes the effect of each preprocessing stage applied sequentially to the raw corpus. The cleaning process was conducted in multiple stages. First, entries containing empty strings, null values, or punctuation-only text were removed to eliminate non-informative content introduced during data extraction and merging.

Next, duplicate sentence entries arising from cross-source aggregation were removed to reduce redundancy in the corpus. A length-based filtering step was then applied to exclude sentences that were either excessively short or overly long. Sentences containing fewer than two words were removed due to insufficient contextual information. In comparison, sentences exceeding thirteen words were excluded to avoid verse-level or narrative-style sequences that are less suitable for sentence-level translation modeling. For example, verse-based entries such as Genesis 4:8, which contain multiple clauses and narrative actions within a single unit, frequently exceeded the length threshold and were therefore excluded. This step accounted for the most significant reduction in corpus size, reflecting the heterogeneous nature of the raw data.

Finally, alignment quality was enforced using a length ratio constraint of 1.33. Sentence pairs exceeding this ratio were treated as poorly aligned and removed, as such cases typically indicate partial translations or mismatched sentence boundaries rather than valid parallel sentences. For example, an Ilocano sentence containing six words paired with a Tagalog sentence containing more than eight words often reflected added clauses or incomplete correspondence, resulting in a length ratio beyond the acceptable range.

After these primary preprocessing stages, additional consolidation and validation procedures were applied to resolve residual cross-source overlaps and inconsistencies in alignment. These final checks yielded a dataset of 10,038 parallel sentence pairs, which was used for model training and evaluation.

Dataset Distribution, Split Integrity, and Leakage-Free Confirmation

Following data cleaning and alignment, the finalized Ilocano-Tagalog parallel corpus consisted of 10,038 sentence pairs. To ensure reliable model evaluation and leakage-free dataset handling, the corpus was divided into training, validation, and test sets using a two-step holdout strategy applied prior to any data augmentation or model training.

First, 10% of the corpus (n = 1,004) was reserved as a held-out test set and excluded from all training and tuning procedures. The remaining 90% (n = 9,034) was further split into training and validation subsets using a 90/10 ratio, yielding 8,130 training pairs and 904 validation pairs. This resulted in an approximate 81% / 9% / 10% distribution for training, validation, and testing, respectively.

Table 7. Final Dataset Distribution and Train-Validation-Test Split

Dataset Partition	Number of Sentence Pairs	Percentage (%)	Purpose
Training	8,130	81.0%	Model fine-tuning and data augmentation
Validation	904	9.0%	Model monitoring and parameter evaluation
Test	1,004	10.0%	Final performance evaluation
Total	10,038	100%	-

Throughout the experimental pipeline, strict separation between training, validation, and test sets was enforced to confirm leakage-free evaluation. The training set was used exclusively for model fine-tuning and mask-based data augmentation, while the validation set was used solely for model monitoring and intermediate evaluation. The test set was retained in its original form and accessed only for final performance assessment.

As part of the leakage-free dataset confirmation, mask-based data augmentation was applied exclusively to the training partition, expanding the dataset from 8,130 to 32,520 sentence pairs. The validation (n = 904) and test (n = 1,004) sets were kept fixed and were never augmented, duplicated, or modified at any stage of preprocessing or training. All automatic evaluation metrics, including BLEU, ChrF, and METEOR, as well as human evaluation scores, were computed solely on the held-out validation and test sets.

This dataset organization preserves split integrity and ensures that reported results generalize to unseen data, rather than relying on memorization or exposure to augmented training instances. Given the unusually high automatic evaluation scores observed later in this chapter, additional safeguards were applied to confirm dataset integrity. Sentence-level hashing and string-matching checks were performed to ensure that no exact or near-duplicate source-target pairs occurred across training, validation, and test partitions.

Statistics of the Ilocano-Tagalog Parallel Corpus

Table 8 presents the descriptive statistics of the final aligned corpus. Both languages contain 10,038 sentences, with the shortest sentence at two

words and the longest at 13 words. On average, Ilocano sentences are slightly longer (6.59 words) than Tagalog sentences (6.36 words). The Ilocano dataset also has more unique words (2,699) compared to Tagalog (2,473), reflecting greater lexical diversity and variation in expression.

Table 8. Descriptive Statistics of the Final Ilocano-Tagalog Parallel Corpus

Descriptive Statistics	Ilocano	Tagalog
Number of sentences	10038	10038
Number of Words Overall	66190	63889
Total Number of Unique Words	2699	2473
Length of Shortest Sentence	2	2
Shortest Sentence	kumusta ka	kumusta ka
Length of Longest Sentence	13	13
Longest Sentence	adda kadi ti plano yo a makipag istoria iti gagayyem iti Saturday night	may plano ba kayo na makipag usap sa kaibigan sa sabado ng gabi
Average Word Count in a Sentence	6.59	6.36

In terms of vocabulary, the Ilocano dataset contains significantly more unique words (2,699) than Tagalog (2,473). This suggests that Ilocano exhibits greater word variation, possibly due to richer morphology or a more diverse lexicon. The higher word count in Ilocano (66,190 vs. 63,889) also reflects this difference. This lexical diversity may pose additional challenges for machine translation models, requiring more sophisticated handling of inflections and synonyms.

Sentence Length Distribution

The sentence-length distribution of the Ilocano-Tagalog parallel corpus, illustrated in Figures 3 and 4, is concentrated between four and eight words, with the highest frequency at approximately five words per sentence (over 2,400 instances). The overall pattern is roughly bell-shaped, with frequencies

declining sharply for very short sentences (below four words) and for longer sentences beyond nine to ten words. This indicates that the corpus is primarily composed of short-to-medium-length sentences, a structure commonly associated with conversational and instructional texts.

The two languages exhibit highly similar sentence length distributions, indicating strong structural alignment between source and target sentences. Minor differences are nevertheless observed: Ilocano shows a slight predominance in the eight-ten-word range. At the same time, Tagalog demonstrates higher frequencies in around six-word sentences, reflecting expected morphological and syntactic variation between the languages.

Overall, the consistency in sentence length across the parallel corpus supports its suitability for sentence-level machine translation. However, the limited representation of longer and syntactically complex sentences suggests that translation performance may be more constrained for extended constructions.

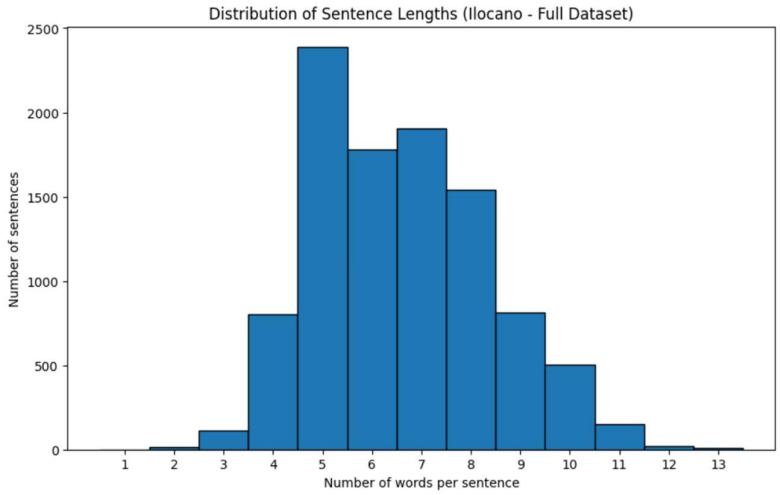


Figure 3. Distribution of Tagalog and Ilocano Sentences Word Count

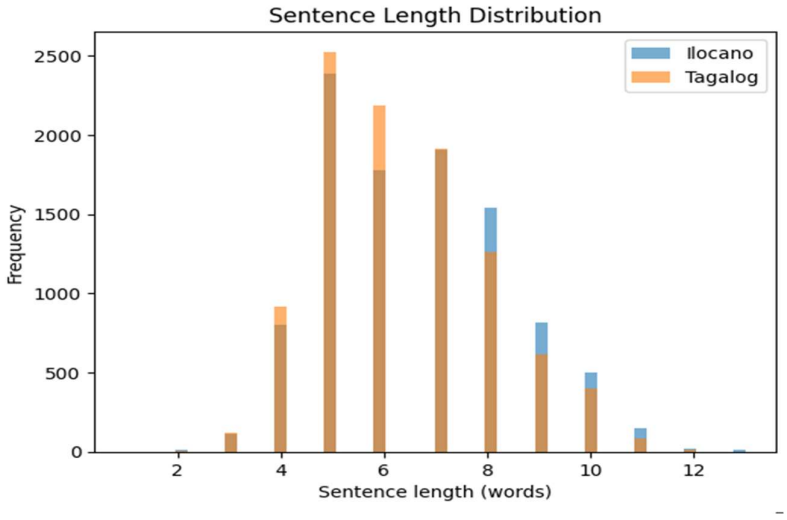


Figure 4. Sentence Length Distribution

Source-Target Length Correlation

The correlation between Ilocano (source) and Tagalog (target) sentence lengths in the parallel corpus is portrayed in Figure 5. The scatterplot demonstrates a strong positive correlation: longer Ilocano sentences generally correspond to longer Tagalog sentences, while shorter Ilocano sentences correspond to shorter Tagalog sentences. Most data points lie close to the diagonal, indicating strong alignment between the two languages.

In a few cases, Ilocano sentences of the same length corresponded to Tagalog sentences of varying lengths. This variation can be attributed to morphological, syntactic, or lexical differences, as Tagalog may use either more or fewer words to express the same meaning. Such differences are expected, and overall, the distribution demonstrates strong sentence-level alignment, which is essential for practical training and evaluation of machine translation models.

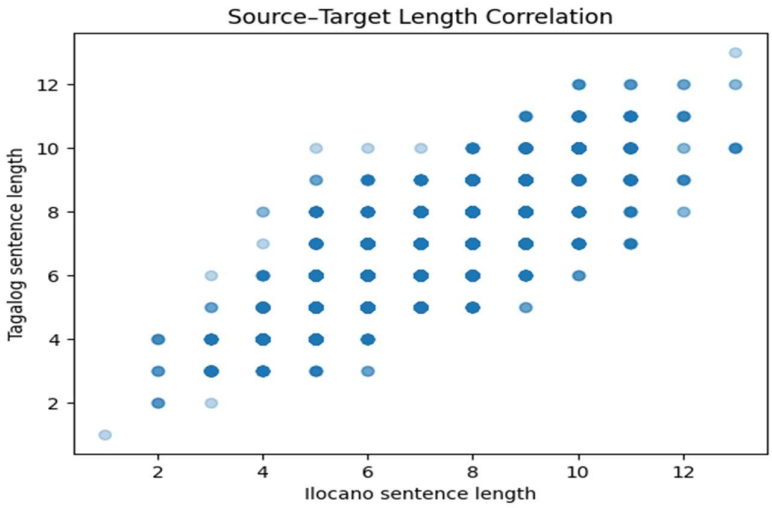


Figure 5. Source-Target Length Distribution

Word Frequency Distribution and Zipf's Law

The word frequency distribution of the Ilocano corpus follows a common pattern observed in natural language, where a small set of words appears very frequently. In contrast, most words occur only a few times. Common function words and particles are used repeatedly across many sentences, whereas many content words appear infrequently, with some occurring only once or twice in the entire dataset. This imbalance between frequent and rare words is a normal characteristic of human language.

This pattern is illustrated in Figure 6, which shows a steep decline in word frequency as word rank increases, consistent with Zipf's Law. From a machine translation perspective, the presence of many low-frequency words can make learning more difficult because the model has limited exposure to them during training. However, the fact that the Ilocano corpus follows this expected Zipfian distribution indicates that the language reflects natural usage and is appropriate for training and evaluating translation models.

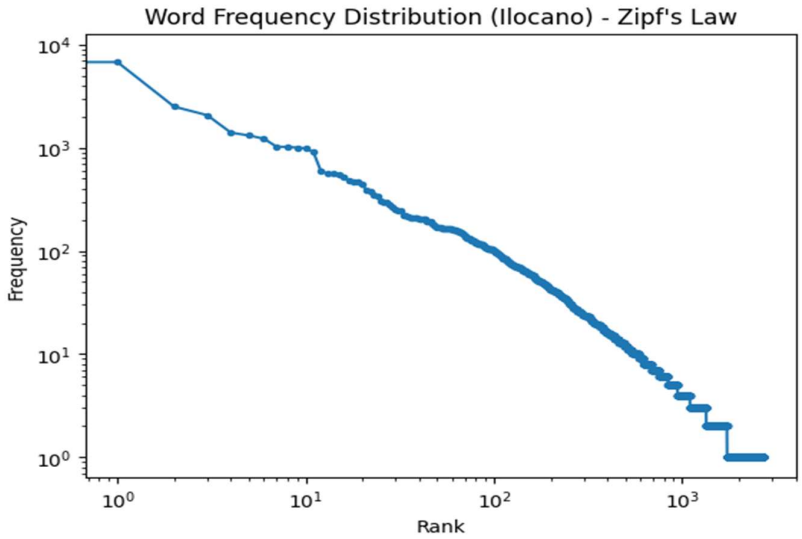


Figure 6. Word Frequency Distribution (Ilocano) - Zipf's Law

Data Augmentation and Tokenization Analysis

This section examines the application of mask-based data augmentation and its effects on training data size and token-level characteristics relevant to model learning. The analysis focuses on how augmentation alters the training corpus while preserving the integrity of the validation and test sets.

The cleaned Ilocano-Tagalog dataset was loaded from an Excel file, converted into a Hugging Face Dataset object, and standardized by normalizing column names to ensure consistent field access. The corpus was then split using a two-step holdout procedure with a fixed random seed (seed = 42), resulting in an approximate 81% training, 9% validation, and 10% test split. This partitioning was completed prior to augmentation, and the validation and test sets were kept unchanged throughout the experiment.

Mask-based data augmentation was applied exclusively to the training set. Ilocano source sentences were tokenized at the word level using whitespace segmentation, after which each word was replaced with a special

mask token (<mask>) with probability 15%. The masked sentence served as a new source of input, while the corresponding Tagalog target sentence remained unmodified. This strategy introduces controlled noise into the source text while preserving correct target translations, encouraging the model to learn stronger contextual and lexical representations.

A short ablation analysis was conducted to examine the effect of different mask rates on validation performance and to guide the selection of an appropriate augmentation configuration. Three masking probabilities (10%, 15%, and 30%) were evaluated under identical training budgets using validation BLEU as the selection criterion. Based on this observation, a 15% masking rate was selected as a balanced configuration, providing sufficient augmentation-induced variability without excessively degrading validation performance.

Table 9. Validation BLEU under Different Mask Rates (Short Ablation)

Mask Rate	Training Budget	Validation BLEU
10%	600 steps	78.60
15% (used)	600 steps	75.85
30%	600 steps	72.72

For further increase in training diversity, multiple masked variants were generated for each training example. Specifically, four masked versions per sentence were created, expanding the training set from 8,130 to 32,520 sentence pairs. No augmentation, duplication, or modification was applied to the validation or test sets, ensuring that all evaluation results reflect performance on unaltered data. This augmentation strategy helps the model learn to handle incomplete or missing information, improving sustainability in real-world scenarios. Exposing the model to varied masked inputs can help it better generalize across different sentence structures and lexical patterns. The

approach also mitigates overfitting by increasing the adequate size of the training data without introducing noise into evaluation sets. Such targeted augmentation is particularly beneficial for low-resource languages, where the amount of original parallel data is limited. Overall, this method strengthens the model's ability to predict accurate translations even when presented with unfamiliar or partially obscured inputs. Additionally, this encourages the model to capture contextual dependencies more effectively, enhancing translation quality. The multiple masked variants also serve as a form of regularization, improving model stability during training. Finally, this strategy provides a scalable way to expand datasets without requiring additional manual annotation. Moreover, this helps the model become more resilient to variations in sentence length and structure. This also facilitates learning subtle linguistic patterns that might otherwise be underrepresented in small datasets. This approach ultimately contributes to creating a more effective and adaptable neural machine translation system for low-resource languages.

Table 10. Summary of Dataset Distribution Before and After Augmentation

Dataset Split	Before Augmentation	After Augmentation
Training	8,130	32,520
Validation	904	904
Test	1,004	1,004
Total	10,038	34,428

Table 10 illustrates that the overall dataset size increased substantially due to augmentation, with growth occurring exclusively in the training partition. By keeping the validation and test sets unchanged, the experimental setup ensures that performance comparisons between augmented and non-augmented models remain fair and unbiased.

Tokenization Statistics of Augmented and Non-Augmented Data

Word-level statistics describe corpus composition, while token-level statistics reflect model input complexity after subword segmentation and are therefore not directly comparable. Tokenization statistics were computed using the MarianMT SentencePiece tokenizer to examine how mask-based augmentation affects sequence length and token distribution in the training data, as summarized in Table 11.

Table 11. Tokenization Statistics of Augmented and Non-Augmented Training Data

Model Variant	Training Sentences	Avg Tokens / Sentence	Median Token	Min Token	Max Tokens
Non-Augmented	8,130	13.90	14	3	29
Augmented	32,520	15.93	15	3	37

The augmented training set exhibits higher average and maximum token counts per sentence compared to the non-augmented set, reflecting the introduction of masked tokens that may be segmented into multiple subword units during tokenization. Despite this increase, the minimum token count remains unchanged, indicating that the augmentation process does not introduce empty or malformed sequences.

These token-level differences indicate that mask-based augmentation exposes the model to longer and more diverse token sequences during training. Such exposure may support improved handling of contextual variation and low-frequency lexical items during translation, while preserving structural consistency between augmented and original training data.

Model Training

The training configuration used to fine-tune the MarianMT model for the Ilocano-Tagalog translation task is summarized in Table 12. The pretrained MarianMT model (Helsinki-NLP/opus-mt-tl-en) was adapted for Ilocano-Tagalog translation through supervised fine-tuning. Both the baseline (non-augmented) and augmented models employed identical architectures, optimizers, and hyperparameter settings to ensure a fair and controlled comparison. Training was conducted using the AdamW optimizer with an initial learning rate of 5.00×10^{-5}, a mini-batch size of 32, and a linear learning rate decay scheduler. Each model was fine-tuned for five epochs, with evaluation performed at the end of every epoch to monitor learning progress and reduce the risk of overfitting.

Due to the increased size of the augmented training set, the augmented model required substantially longer training time (approximately 99 hours) compared to the baseline model (approximately 40 hours). Despite this difference in training duration, evaluation procedures remained consistent across both models, ensuring that observed performance differences reflect learning behavior rather than bias in training configuration.

Table 12. Training Setup for Controlled Comparison Between Model Variants

Parameter	Non-Augmented	Augmented
Model	MarianMT (Helsinki-NLP/opus-mt-en-tl)	MarianMT (Helsinki-NLP/opus-mt-en-tl)
Evaluation Strategy	End of every epoch	End of every epoch
Training Data Size	8,130 sentence pairs	32,520 sentence pairs
Training Duration	~40 hours	~99 hours

Automatic Evaluation Results

Table 13 presents the automatic evaluation results of the MarianMT models on 1,004 Ilocano-Tagalog sentence pairs. Translation quality is measured using BLEU, ChrF, and METEOR, which capture word-level overlap, character-level similarity, and semantic correspondence with human reference translations.

The augmented MarianMT model achieves higher scores across all metrics, with improvements observed in BLEU (88.24 vs. 81.93), ChrF (92.86 vs. 89.25), and METEOR (0.9284 vs. 0.8884). These gains indicate that mask-based augmentation leads to closer lexical alignment, improved character-level consistency, and stronger semantic matching with reference translations. Although the non-augmented model performs competitively, the consistently lower scores suggest reduced coverage of lexical variation and contextual patterns.

Overall, the results indicate that data augmentation improves translation quality rather than relying on memorization, as performance gains are observed on data not used during training. These results suggest that mask-based data augmentation primarily enhances lexical and character-level alignment with reference translations, likely by exposing the model to a broader range of contextual variations during training rather than encouraging memorization of fixed sentence patterns.

The results reported in Table 13 include evaluation under both the original test split and an additional strict no-overlap condition designed to rule out sentence-level overlap, with corresponding generalization results detailed later in Table 24.

Table 13. Overall Test Set Performance

Model Variant	BLEU	ChrF	METEOR	N
MarianMT (Augmented)	88.24	92.86	0.9284	1004
MarianMT (Non-Augmented)	81.93	89.25	0.8884	1004

Although the reported BLEU, ChrF, and METEOR scores are high compared with many low-resource translation benchmarks, these values should be interpreted in relation to the dataset characteristics and evaluation setup used in this study. The parallel corpus was aggregated from multiple sources and filtered to retain sentence pairs from selected domains characterized by relatively limited syntactic variability and restrained idiomatic usage. Such conditions are known to yield higher n-gram overlap between model outputs and reference translations, particularly for closely related Philippine languages with lexical and structural similarity.

Augmented vs Non-Augmented Outputs

Table 14. Output-level comparison of augmented and non-augmented models.

Metric	Augmented	Non-Augmented	Δ
BLEU	88.24	81.93	+6.31
ChrF	92.86	89.25	+3.61
METEOR	0.9284	0.8884	+0.0400

Table 14 provides a direct output-level comparison between the augmented and non-augmented MarianMT models by reporting the absolute differences (Δ) in BLEU, ChrF, and METEOR scores computed on the same test set. Unlike Table 14, which presents the overall performance of each model, this table highlights the magnitude of improvement attributable to mask-based data augmentation. Consistent gains are observed across all evaluation metrics. The augmented model improves BLEU by 6.31 points, ChrF by 3.61 points, and

METEOR by 0.0400 relative to the non-augmented model. These increases indicate that augmentation yields systematically better alignment with reference translations at the word, character, and semantic levels. Rather than reflecting isolated metric fluctuations, the differences summarized in this table show a uniform advantage for the augmented model under identical evaluation conditions, reinforcing the contribution of mask-based augmentation to improved translation output quality. This demonstrates that strategic data augmentation can consistently enhance model performance across multiple evaluation metrics.

Sentence Length-Based Results

Translation performance varies systematically with sentence length, providing insight into how each model handles increasing sequence complexity. Sentence pairs were grouped into short, medium, and long categories, and evaluated using BLEU, ChrF, and METEOR for both augmented and non-augmented MarianMT models.

For short sentences, both models achieve strong performance, reflecting the relative simplicity and high frequency of shorter constructions in the corpus. The augmented model nonetheless attains higher scores across all metrics, indicating more accurate alignment with reference translations even for simpler inputs. In the medium-length category, performance declines for both models as sentence structure becomes more complex. Despite this increase in difficulty, the augmented model consistently maintains higher BLEU, ChrF, and METEOR values than the non-augmented variant. Performance increases again for long sentences in both models, with the augmented model achieving

the highest scores in this category. This pattern may be influenced by the smaller number of long sentences and their relatively regular syntactic structure within the dataset.

Across all sentence-length categories, the augmented model demonstrates more stable and accurate output-level translations, suggesting that augmentation benefits extend beyond overall averages and persist across varying sequence lengths. The consistent advantage of the augmented model indicates that augmentation benefits are not limited to a specific input length range, although performance remains strongest for short and medium sentences.

Table 15. Sentence Length-Based Output Performance

Sentence Length	MarianMT Variant	BLEU	ChrF	METEOR	N
Short	Augmented	88.05	91.49	0.9274	343
	Non-Augmented	80.57	87.71	0.8791	343
Medium	Augmented	86.42	92.08	0.9196	517
	Non-Augmented	79.24	88.19	0.8784	517
Long	Augmented	92.73	96.46	0.9626	144
	Non-Augmented	89.55	93.75	0.9467	144

System-Level Output Comparison

This evaluation compares the outputs of the augmented and non-augmented MarianMT models with those of Google Translate and ChatGPT on a subset of 80 Ilocano sentences. For this comparison, human-produced Tagalog translations of the selected Ilocano sentences were used as reference translations, serving as the target outputs for all systems.

Table 16. System-level comparison using human reference translations

System	BLEU	ChrF	METEOR	N
Augmented	35.67	64.22	0.5966	80
Non-Augmented	35.88	64.46	0.5846	80
Google Translate	26.44	55.39	0.5098	80
ChatGPT	30.94	62.92	0.5888	80

The results shown in Table 16 indicate that both MarianMT variants achieve higher BLEU, ChrF, and METEOR scores than Google Translate and ChatGPT. The augmented MarianMT model attains a BLEU score of 35.67, a ChrF score of 64.22, and a METEOR score of 0.5966, while the non-augmented model yields comparable performance. In contrast, the external systems obtain noticeably lower scores across all metrics.

The lower scores of Google Translate and ChatGPT can be attributed to their general-purpose design. These systems are trained to support many language pairs and domains, whereas Ilocano-Tagalog is a low-resource language pair that is sparsely represented in large-scale multilingual training data. As a result, their outputs may rely on indirect transfer mechanisms, leading to reduced alignment with the human reference translations for this specific task.

In addition, general-purpose systems often generate fluent paraphrases that differ lexically or structurally from the reference translation. While such outputs may be acceptable to human readers, automatic metrics such as BLEU and ChrF favor closer surface-level correspondence, which can lower scores for paraphrased translations.

In contrast, the MarianMT models were fine-tuned directly on an Ilocano-Tagalog parallel corpus drawn from a consistent domain. This targeted training

allows the models to better capture lexical choices, morphological patterns, and translation conventions reflected in human reference translations, resulting in higher automatic evaluation scores.

This comparison highlights the advantage of task-specific fine-tuning for low-resource language pairs, where targeted training enables closer alignment with domain-consistent human references than general-purpose multilingual systems. This also underscores the importance of customizing models to the linguistic characteristics and data constraints of the target language pair. As a result, such fine-tuning can lead to more accurate and contextually appropriate translations in practical applications.

Statistical Significance Testing via Paired Bootstrap Resampling

Statistical significance testing was conducted to verify that the observed BLEU improvement of the augmented MarianMT model is not attributable to random variation in the test data. Paired bootstrap resampling was applied following Koehn (2004), using the complete test set of 1,004 Ilocano-Tagalog sentence pairs.

Statistical significance testing was conducted on sentence-level metric differences rather than on corpus-level aggregates to reduce inflation effects arising from high baseline scores. The procedure repeatedly resampled the test set with replacement while preserving sentence-level alignment between the augmented and non-augmented model outputs. For each resampled set, BLEU scores were recomputed for both models, and the difference between them was recorded. This process was repeated 10,000 times to obtain an empirical distribution of BLEU score differences. Across the resampling trials, the

augmented model achieved an average BLEU score of 74.23, compared to 61.76 for the non-augmented model, resulting in an observed improvement of 12.47 BLEU points. The 95% confidence interval for the BLEU difference was [10.56, 14.41], which does not include zero, and the one-sided p-value was below 0.001. These results indicate that the augmented model consistently outperforms the baseline across resampled versions of the same test data.

The findings provide strong statistical evidence that the performance gains from mask-based data augmentation are stable and unlikely to be due to chance, reinforcing the reliability of the observed improvements in Ilocano-Tagalog translation quality.

Table 17. Paired Bootstrap Resampling Results on the Full Test Set (N = 1,004)

Model Variant	BLEU	ΔBLEU	95% CI for ΔBLEU	One-sided p-value
Non-Augmented	61.76	–	–	–
Augmented	74.23	+12.47	[10.56, 14.41]	< 0.001

Inter-Rater Agreement in Human Evaluation (Weighted Cohen's κ)

Inter-annotator agreement was measured using weighted Cohen's κ and computed separately for the augmented and non-augmented evaluation sets to avoid conflating annotator consistency across conditions. All ratings were based on identical 5-point Likert-scale rubrics and included 195 sentence-level evaluations per set.

As shown in Table 18, agreement values for adequacy, fluency, and faithfulness ranged from low to fair for both model variants. For the augmented model, fluency exhibited the highest agreement (κ = 0.296), while adequacy and faithfulness showed lower agreement, reflecting greater variability in semantic-level judgments. For the non-augmented model, adequacy achieved

the highest agreement (κ = 0.338), whereas fluency showed the lowest agreement (κ = 0.105).

The results indicate that data augmentation affects both translation quality and the consistency of human judgments. Higher fluency agreement in the augmented model suggests more uniform surface-level output, while lower agreement in adequacy and faithfulness reflects greater variability in semantic interpretation. In contrast, the non-augmented model shows more consistent adequacy judgments but lower agreement on fluency, highlighting a trade-off between semantic stability and linguistic naturalness in low-resource machine translation.

Table 18. Inter-Rater Agreement Results Using Weighted Cohen's κ

Model Variant	Dimension	κ (Weighted)	N
Augmented	Adequacy	0.204	195
	Fluency	0.296	195
	Faithfulness	0.230	195
	Overall	0.254	195
Non-Augmented	Adequacy	0.338	195
	Fluency	0.105	195
	Faithfulness	0.243	195
	Overall	0.242	195

Human Validation Results for Augmented and Non-Augmented Models

Human evaluation results, summarized in Table 19, indicate consistently high sentence-level translation quality for both model variants. Mean ratings exceed 4.4 out of 5 across adequacy, fluency, faithfulness, and overall quality, indicating that translations produced by both systems are generally perceived as acceptable to very good. The uniformly high scores across all dimensions suggest that neither model exhibits systematic degradation in sentence-level output quality.

Apparent differences nevertheless emerge between the two training strategies. As shown in Table 19, the non-augmented model achieves slightly higher mean scores and a greater proportion of ratings at or above the ≥ four threshold, particularly for fluency and overall quality, while also exhibiting lower standard deviations. This pattern indicates more consistent sentence-level judgments, reflecting smoother phrasing and more predictable surface realization. In contrast, the augmented model records marginally lower mean scores, with higher variability across all dimensions.

Mean ratings for the augmented model range from 4.44 to 4.50, with standard deviations between 0.69 and 0.73, suggesting strong overall quality accompanied by greater dispersion in evaluator judgments. This increased variability is consistent with paraphrastic variation and less uniform lexical choices introduced during augmentation.

Table 19. Summary of Human Evaluation Scores Across Evaluation Dimensions

Model Variant	Dimension	Mean	SD	≥4 (%)	N
Augmented	Adequacy	4.50	0.69	88.21	195
	Fluency	4.44	0.73	88.21	195
	Faithfulness	4.46	0.72	88.21	195
	Overall	4.47	0.71	88.21	195
Non-Augmented	Adequacy	4.62	0.58	90.77	195
	Fluency	4.76	0.44	95.90	195
	Faithfulness	4.53	0.73	87.18	195
	Overall	4.63	0.50	93.33	195

Taken together, the results in Table 19 suggest a trade-off between output consistency and lexical variability. While the non-augmented model benefits from more uniform fluency and surface-level smoothness, the augmented model produces outputs that remain high in perceived quality but

exhibit greater diversity in sentence realization. This distinction highlights how data augmentation influences human-perceived translation characteristics without substantially degrading overall quality, particularly in terms of expressive flexibility and adaptability across varied linguistic contexts and evaluation conditions. As such, the augmented approach demonstrates a favorable balance between maintaining translation adequacy and encouraging natural linguistic variation**.**

Error Annotation Analysis

Tables 20 and 21 summarize the frequency and distribution of translation errors identified in 195 evaluated Ilocano-Tagalog sentence pairs for the augmented and non-augmented models. Errors were categorized into mistranslation, omission, addition, word-order error, grammar, and term inconsistency, with counts and percentages reported as a proportion of the total number of evaluated sentences.

For the augmented model, mistranslation errors are the most frequent (6.15%), followed by grammatical (1.54%) and omission errors (1.03%). No additional or word order errors were observed, and term inconsistency occurs only once (0.51%). In contrast, the non-augmented model shows higher error frequencies overall, particularly for mistranslation (9.23%) and grammatical errors (5.13%), with addition and word order errors appearing only in this variant. Omission errors occur at the same rate in both models.

Taken together, these results indicate that the augmented model produces fewer semantic and grammatical errors than the non-augmented model, while both systems exhibit similar omission behavior. The error

distributions provide qualitative insight into the types of translation issues present in each model and complement the automatic evaluation results.

Table 20. Error Annotation Frequency (Non-Augmented Model)

Error Type	Count	% of 195
Mistranslation	18	9.23
Omission	2	1.03
Addition	2	1.03
Word order error	1	0.51
Grammar	10	5.13
Term inconsistency	0	0.00

Table 21. Error Annotation Frequency (Augmented Model)

Error Type	Count	% of 195
Mistranslation	12	6.15
Omission	2	1.03
Addition	0	0.00
Word order error	0	0.00
Grammar	3	1.54
Term inconsistency	1	0.51

Model Validation

This section examines the performance of the fine-tuned MarianMT models from multiple evaluation perspectives, including generalization across data conditions, training behavior, output structural characteristics, and comparative analysis against reference systems. The goal is to assess model behavior and reliability beyond surface-level output scores. This approach supports a more comprehensive interpretation of translation quality in a low-resource setting.

Training Performance Across Epochs

Tables 22 and 23 summarize the training and evaluation metrics recorded at the end of each fine-tuning epoch for the non-augmented and augmented MarianMT models, respectively. Reported metrics include training

loss, evaluation loss, BLEU score, and learning rate, providing a concise numerical overview of model learning behavior across five epochs.

For the non-augmented model (Table 22), training and evaluation loss decrease consistently across epochs, indicating stable optimization and effective convergence. BLEU scores improve steadily from the first to the fourth epoch, then stabilize in the final epoch. The learning rate follows a linear decay schedule, gradually decreasing toward zero, which supports controlled parameter updates during later training stages.

Table 22. Training and Evaluation Metrics Across Epochs (Non-Augmented Model)

Epoch	Train Loss	Eval Loss	BLEU	Learning Rate
1	0.2721	0.0531	65.76	4.00×10^{-5}
2	0.0456	0.0361	77.85	3.00×10^{-5}
3	0.0310	0.0305	81.69	2.00×10^{-5}
4	0.0243	0.0286	82.87	$1.00\times1\ 0^{-5}$
5	0.0212	0.0262	82.02	~0

A similar training pattern is observed for the augmented model (Table 23), with both training and evaluation loss exhibiting a smooth downward trend across epochs. BLEU scores increase more rapidly in the early epochs and stabilize at a higher level than those of the non-augmented model. The learning rate schedule mirrors that of the non-augmented model, confirming that both models were trained under identical optimization settings.

Table 23. Training and Evaluation Metrics Across Epochs (Augmented Model)

Epoch	Train Loss	Eval Loss	BLEU	Learning Rate
1	0.0273	0.0531	82.32	4.50×10^{-5}
2	0.0221	0.0361	85.37	3.00×10^{-5}
3	0.0201	0.0305	87.81	2.10×10^{-5}
4	0.0196	0.0286	88.25	1.10×10^{-5}
5	0.0189	0.0262	88.24	~0

Overall, the epoch-level metrics indicate stable training dynamics for both models and align with the trends observed in the TensorBoard training logs. The augmented model's higher and more stable BLEU scores suggest that the expanded training data contributes to improved learning behavior without introducing instability during optimization.

TensorBoard Training and Evaluation Logs

Figures 7 and 8 present the TensorBoard training and evaluation logs for the non-augmented and augmented MarianMT models, respectively. Each figure consists of four subplots showing (a) evaluation BLEU, (b) evaluation loss, (c) learning rate across training steps, and (d) training loss. Across both models, BLEU evaluation increases steadily during training, while training loss and evaluation loss decrease consistently, indicating effective learning and convergence during fine-tuning. The learning rate plots confirm the use of a linear decay scheduler, with the learning rate gradually decreasing toward zero as training progresses, contributing to stable optimization in later training steps.

Comparatively, the augmented model exhibits smoother evaluation curves, particularly in evaluation loss and BLEU progression, which suggests more stable learning behavior across training steps. The smoother curves also highlight the benefit of exposing the model to a more diverse set of training examples. Overall, the TensorBoard logs provide visual confirmation of the stable training dynamics reflected in the epoch-level metrics reported in Tables 22 and 23.

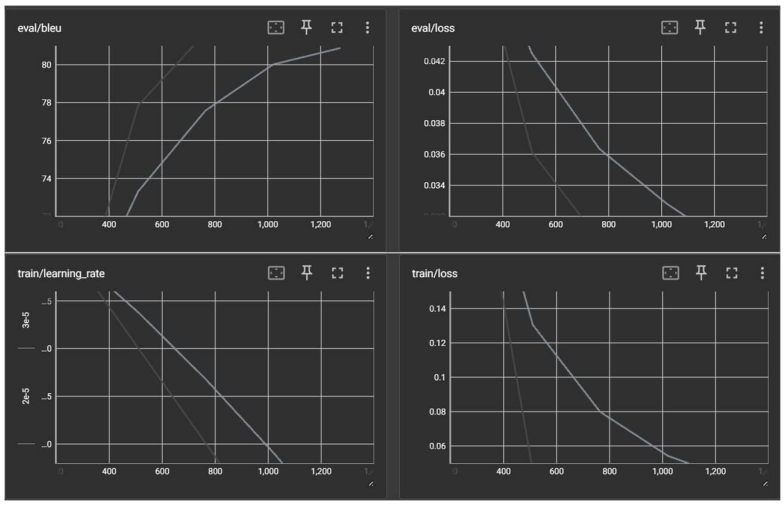


Figure 7. TensorBoard Metrics (Non-Augmented MarianMT): (a) evaluation BLEU, (b) evaluation loss, (c) learning rate across training steps, and (d) training loss

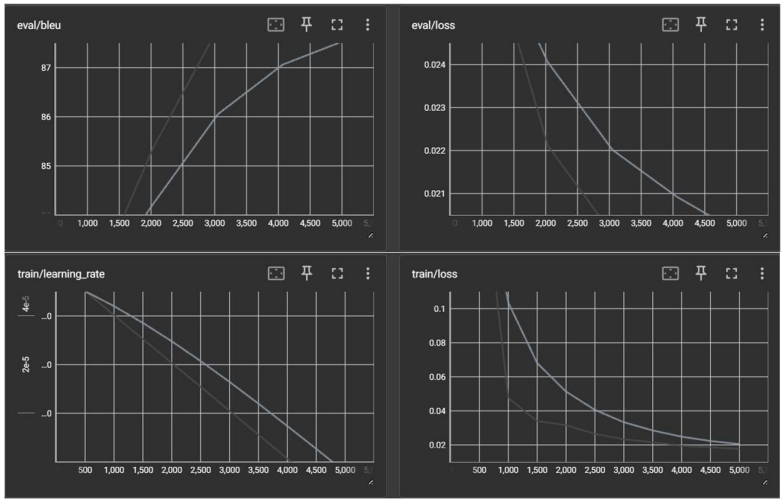


Figure 8. TensorBoard Metrics (Augmented MarianMT): (a) evaluation BLEU, (b) evaluation loss, (c) learning rate across training steps, and (d) training loss.

Generalization Evaluation

Generalization was assessed using a strict no-overlap evaluation condition to examine whether the proposed MarianMT model relied on reusable translation patterns rather than sentence-level reuse from the training data. This evaluation setting was designed to reduce the influence of memorized sentence

pairs and to provide a conservative assessment of model behavior under controlled dataset conditions. Under the strict no-overlap evaluation, the test set was restricted to sentence pairs that were not lexically identical or near-identical to any training sentence at the sentence level. Lexical similarity in this context refers exclusively to sentence-level identity or near-identity and does not exclude shared vocabulary, subword units, or commonly occurring linguistic constructions.

As a result, the evaluation enforces sentence-level independence while preserving the same sentence style, length, and level of syntactic and idiomatic complexity as in the original dataset. The strict no-overlap condition allows examination of translation performance on sentence instances not encountered during training while remaining within the same linguistic conditions defined by the dataset.

Table 24. Generalization Performance under Original and Strict No-Overlap Evaluation Conditions

Model Variant	Evaluation Condition	BLEU	ChrF	METEOR	No. Pairs
Augmented	Original Test Set	88.24	92.86	0.928	1,004
Augmented	Strict No-Overlap	85.06	91.10	0.906	692
Non-Augmented	Original Test Set	82.02	89.31	0.890	1,004
Non-Augmented	Strict No-Overlap	78.12	87.11	0.860	692

Due to the enforcement of sentence-level independence, the number of evaluated sentence pairs decreased from 1,004 in the original test set to 692 in the strict no-overlap evaluation. For the augmented MarianMT model, BLEU decreased from 88.24 under the original test condition to 85.06 under the overlap-controlled setting, while ChrF declined from 92.86 to 91.10 and METEOR from 0.928 to 0.906. Similarly, the non-augmented model exhibited

reductions in BLEU from 82.02 to 78.12, ChrF from 89.31 to 87.11, and METEOR from 0.890 to 0.860 when evaluated under the same conditions.

Minor numerical differences in reported metric values may arise from metric computation settings or rounding precision and do not affect the observed comparative trends.

Despite increased difficulty imposed by the strict no-overlap constraint, both models maintained high automatic evaluation scores across all metrics. The augmented model showed a BLEU reduction of approximately 3.2 points, while the non-augmented model exhibited a slightly larger decrease of approximately 3.9 points. A similar magnitude of performance degradation across models indicates that neither system relies primarily on sentence-level memorization to achieve strong automatic scores.

Collectively, these findings indicate that both MarianMT-based models can maintain translation quality on sentence instances not encountered during training under the same dataset conditions. The augmented model demonstrated more stable performance under overlap-controlled evaluation, suggesting that mask-based data augmentation contributes to greater consistency when sentence reuse is removed.

Overall, the results support sentence-level generalization for Ilocano–Tagalog machine translation within the experimental constraints of this study, while clarifying that the evaluation does not address changes in language use, stylistic register, or topical variation beyond those represented in the dataset.

CHAPTER 5

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

Summary of the Study

Machine translation for low-resource language pairs remains challenging due to limited parallel data and uneven domain coverage. This study examined whether mask-based data augmentation and controlled validation strategies can improve neural machine translation performance for Ilocano-Tagalog, a low-resource Philippine language pair. Two MarianMT-based models were developed and evaluated: a non-augmented baseline model and an augmented model trained with mask-based data augmentation under identical training configurations.

A parallel corpus was constructed by combining publicly available resources and researcher-generated sentence pairs, resulting in 42,273 raw entries. After systematic cleaning, alignment, and validation, a high-quality dataset of 10,038 parallel sentence pairs was retained, consisting primarily of short, conversational, and structurally simple sentences drawn from a limited domain. The dataset was split into training, validation, and test sets in an 81-9-10 ratio. Data augmentation was applied exclusively to the training set, expanding the effective training data from 8,130 to 32,520 sentence pairs by using masked variants that preserve the original semantic content, with a moderate masking rate selected based on a brief ablation analysis and downstream evaluation.

Automatic evaluation on the held-out test set of 1,004 sentence pairs showed consistent and substantial improvements for the augmented model.

The augmented MarianMT model achieved a BLEU score of 88.24, a ChrF score of 92.86, and a METEOR score of 0.9284, outperforming the non-augmented model in a controlled evaluation setting characterized by short-to-medium-length sentences, limited domain variation, and close linguistic similarity between Ilocano and Tagalog. These results indicate that mask-based augmentation improved lexical alignment, character-level consistency, and semantic correspondence with reference translations. Performance gains were consistent across sentence lengths represented in the dataset. For short, medium, and long sentences, the augmented model achieved higher BLEU scores than the baseline, with the most significant gains observed for short and medium-length inputs, which dominate the dataset. This pattern suggests that the benefits of augmentation are not limited to a specific sentence length and extend across varying levels of input complexity.

System-level comparison using 80 Ilocano sentences and human-produced Tagalog reference translations showed that both MarianMT variants outperformed general-purpose translation systems. The augmented model achieved BLEU 35.67, ChrF 64.22, and METEOR 0.5966, while Google Translate and ChatGPT produced substantially lower scores. These findings illustrate the effectiveness of task-specific fine-tuning under controlled experimental conditions for low-resource language pairs, relative to general-purpose multilingual systems not optimized for Ilocano-Tagalog translation.

Model validation analyses further confirmed the reliability of the observed improvements. Paired bootstrap resampling over the whole test set showed a mean BLEU improvement of 12.47 points for the augmented model, with a 95 percent confidence interval from 10.56 to 14.41 and a p-value below

0.001, indicating that the gains are statistically significant. Under a strict no-overlap evaluation condition that reduced the test set to 692 sentence pairs, both the augmented and non-augmented models maintained strong performance despite moderate declines in BLEU, ChrF, and METEOR scores. This consistent behavior across model variants indicates that translation quality is not driven by sentence-level memorization and suggests controlled generalization within the domain and sentence structures represented in the dataset.

Human evaluation on 195 test sentence pairs confirmed high translation quality for both models. The augmented model received mean scores above 4.4 across adequacy, fluency, faithfulness, and overall quality, with over 88 percent of ratings at or above the acceptable threshold. Although the non-augmented model achieved slightly higher fluency ratings, this difference is likely attributable to more conservative and literal output patterns, whereas the augmented model exhibited greater lexical diversity.

Overall, the study demonstrates that source-side mask-based data augmentation, when applied at a moderate level, improves translation accuracy, reduces the frequency of common error types, and supports generalization in Ilocano-Tagalog neural machine translation. By introducing controlled lexical variation during training while preserving target-language integrity, the approach encourages the model to rely on contextual information rather than surface-level sentence patterns. The converging evidence from automatic evaluation metrics, statistical significance testing, and human-assisted evaluation indicates that the proposed augmentation strategy contributes to more reliable and consistent translation performance in low-resource settings.

Conclusions

This section presents the conclusions of the study, organized according to the stated research objectives. Each objective is addressed systematically to interpret the findings and confirm alignment with the research goals.

1. Gather and preprocess an Ilocano-Tagalog parallel corpus, including systematic data collection, cleaning, normalization, and controlled mask-based data augmentation.

This study successfully gathered and curated an Ilocano-Tagalog parallel corpus from multiple publicly available repositories and researcher-constructed sentence pairs, yielding an initial total of 42,273 raw entries. Through systematic preprocessing, comprising the removal of empty and duplicate entries, sentence-length filtering, and alignment validation, the corpus was refined to 10,038 high-quality parallel sentence pairs suitable for neural machine translation. The final dataset exhibited strong structural balance between source and target languages, with identical sentence counts, comparable average sentence lengths (6.59 words for Ilocano and 6.36 words for Tagalog), and consistent length alignment. Vocabulary analysis revealed greater lexical diversity on the Ilocano side (2,699 unique words) than on the Tagalog side (2,473 unique words), reflecting expected morphological variation.

Controlled mask-based data augmentation was applied exclusively to the training split, expanding the training data from 8,130 to 32,520 sentence pairs while keeping the validation (904) and test (1,004) sets unchanged. This preprocessing and augmentation pipeline ensures dataset integrity, leakage-free evaluation, and fair comparison between augmented and non-augmented

models. The results confirm that rigorous corpus preparation is a critical foundation for low-resource neural machine translation.

2. Fine-tune MarianMT-based translation models using augmented and non-augmented datasets to examine the effect of data augmentation on model learning.

The study successfully fine-tuned MarianMT-based translation models using both augmented and non-augmented datasets under identical architectures and hyperparameter configurations. Training dynamics observed across five epochs demonstrated stable convergence for both models, with consistently decreasing training and evaluation loss.

The augmented model exhibited smoother learning curves and achieved higher evaluation of BLEU scores earlier in training, stabilizing at 88.24 BLEU, compared to 82.02 BLEU for the non-augmented model. TensorBoard logs further confirmed stable optimization behavior and effective use of linear learning rate decay in both configurations. A short ablation study on masking intensity showed that a 15% mask rate achieved a higher validation BLEU score (75.85) than a more aggressive 30% mask rate (72.72) under the same training budget, indicating that excessive masking removes critical lexical information. These findings confirm that moderate mask-based augmentation improves learning behavior without introducing training instability.

3. Assess translation output quality using automatic metrics, including BLEU, ChrF, and METEOR, under both human reference and dataset target reference evaluation conditions.

Automatic evaluation on the whole test set (N = 1,004) demonstrated that the augmented MarianMT model consistently outperformed the non-

augmented baseline across all metrics, achieving 88.24 BLEU, 92.86 ChrF, and 0.928 METEOR, compared to 81.93 BLEU, 89.25 ChrF, and 0.888 METEOR for the non-augmented model. The absolute improvements of +6.31 BLEU, +3.61 ChrF, and +0.040 METEOR indicate systematic gains attributable to data augmentation.

To further assess performance under human-reference conditions, a subset of 80 Ilocano sentences was manually translated into Tagalog by a bilingual evaluator as reference translations. Using these references, the augmented MarianMT model achieved 35.67 BLEU, 64.22 ChrF, and 0.5966 METEOR, outperforming Google Translate (26.44 BLEU) and ChatGPT (30.94 BLEU) under the same evaluation setup. These results highlight the advantage of task-specific fine-tuning for low-resource language pairs, where general-purpose systems rely on indirect transfer mechanisms and exhibit weaker alignment with domain-consistent human references.

Across both dataset-target and human-reference evaluation conditions, the augmented MarianMT model demonstrated more substantial lexical alignment, character-level consistency, and semantic correspondence, confirming the effectiveness of mask-based augmentation in improving automatic translation quality.

4. Evaluate model behavior through controlled experimental analysis, including ablation studies, sentence-length-based performance analysis, comparison with benchmark translation systems, human-assisted evaluation using fluency, adequacy, and faithfulness criteria, and statistical analysis of sentence-level performance differences.

Controlled experimental analyses revealed consistent behavioral patterns across multiple evaluation dimensions. Sentence-length-based analysis showed that both models performed best on short and medium-length sentences, which dominate the corpus. In contrast, the augmented model maintained higher BLEU, ChrF, and METEOR scores across all length categories. Under a strict no-overlap generalization condition, test set size was reduced from 1,004 to 692 sentences, yet the augmented model retained high performance (85.06 BLEU), indicating effective generalization beyond sentence-level reuse.

Human-assisted evaluation on 195 sentence pairs showed that both models achieved high perceived translation quality, with mean scores exceeding 4.4 out of 5 across adequacy, fluency, faithfulness, and overall quality. The non-augmented model received slightly higher fluency scores, while the augmented model exhibited greater variability, reflecting increased paraphrastic diversity introduced by augmentation. Error annotation further showed fewer mistranslations and grammatical errors in the augmented model.

Statistical validation using paired bootstrap resampling confirmed that the augmented model's BLEU improvements were statistically significant, with an average gain of 12.47 BLEU points, a 95% confidence interval of [10.56, 14.41], and $p < 0.001$, indicating that observed improvements were not due to random variation.

Taken together, these controlled analyses demonstrate that data augmentation influences not only aggregate scores but also learning behavior, output variability, and generalization capacity. The combined use of automatic metrics, human evaluation, ablation studies, and statistical testing provides a

comprehensive and reliable validation of the proposed Ilocano-Tagalog MarianMT framework.

Recommendations

The following recommendations are presented for future research and the responsible development of low-resource neural machine translation systems for Ilocano-Tagalog and other Philippine languages.

1. Future research should prioritize the continued expansion of Ilocano-Tagalog parallel corpora. Although the curated dataset used in this study enabled strong translation performance, broader coverage is needed to improve generalization. Incorporating additional sources such as conversational transcripts, narrative and literary texts, instructional materials, and domain-specific content may help the model better capture diverse linguistic structures and usage patterns.
2. While mask-based data augmentation demonstrated consistent benefits in this study, future work may explore alternative or complementary augmentation strategies. Techniques such as back-translation, paraphrase generation, synonym substitution, and multilingual augmentation may introduce different forms of lexical and syntactic variation. Comparative studies may help determine whether combining multiple augmentation methods yields further improvements in low-resource translation performance.
3. Human evaluation procedures may be strengthened by involving a larger number of evaluators to improve reliability and reduce subjectivity. In addition to adequacy, fluency, and faithfulness, future evaluations may

include criteria related to stylistic naturalness, pragmatic appropriateness, and cultural sensitivity. Expanding evaluation dimensions can provide a more comprehensive understanding of translation quality from a user-oriented perspective.

4. The experimental framework presented in this study may be extended to multilingual and zero-shot translation scenarios involving other Philippine languages. Investigating cross-lingual transfer among related languages may help assess whether shared linguistic features can improve translation quality in even lower-resource settings and support broader language technology development.
5. Although the proposed Ilocano-Tagalog translation system is presented strictly as a research prototype, future research may examine practical considerations related to real-world use. These include evaluation of inference speed, memory efficiency, and computational requirements, particularly for educational or community-based applications where hardware resources may be limited.
6. Further work may be conducted to conduct more detailed and large-scale error analyses to understand persistent translation issues better. Examining error patterns across different training strategies, sentence lengths, and domains may reveal systematic weaknesses and inform targeted improvements to data preparation, augmentation strategies, or model configurations. Such analysis could also incorporate both automated metrics and human evaluation to capture nuanced translation errors that are not reflected in quantitative scores.

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APPENDICES

APPENDIX A

Letter of Approval

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the capstone project entitled:

"ENHANCING LOW-RESOURCE LANGUAGE PROCESSING: FINE-TUNING MARIANMT FOR ILOCANO – TAGALOG NEURAL MACHINE TRANSLATION"

DANIA MARIE BAUTISTA, REBECCA B. GOMEZ, DANNIEL PAUL B. MOLLENO have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the final presentation.

MARIBEL V. CARDONA
Thesis Project Adviser

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

APPROVAL SHEET

The thesis entitled "**ENHANCING LOW-RESOURCE LANGUAGE PROCESSING: FINE-TUNING MARIANMT FOR ILOCANO - TAGALOG NEURAL MACHINE TRANSLATION**" prepared and submitted by Dania Marie Bautista, Rebecca B. Gomez, and Danniel Paul. B. Molleno in partial fulfillment requirements for the degree Bachelor of Science in Computer Science specialized in Intelligent Systems, is hereby recommended for approval and acceptance.

MARIBEL V. CARDONA
Thesis Project Adviser

Appendix B

Dataset and Experimental Setup Summary

Project Overview

This study investigates Ilocano-Tagalog neural machine translation using MarianMT, comparing a baseline (non-augmented) model with a model trained using mask-based source-side data augmentation.

Dataset Description

- Source Language: Ilocano
- Target Language: Tagalog
- Data Format: Parallel sentence pairs
- Domain: Conversational and instructional text representing everyday language use

Preprocessing Steps

- Column names standardized (lowercase, whitespace removed)
- Removal of:
    - empty or null sentence pairs
    - invalid rows after masking
- All preprocessing performed prior to model training

Train / Validation / Test Split

- Test set: 10% of the dataset (seed = 42)
- Remaining 90% split into:
    - Training set: 90%
    - Validation set: 10%
- Splitting performed using Hugging Face train_test_split

Data Augmentation Method

- **Type:** Source-side stochastic token masking
- **Mask probability:** 0.15
- **Number of variants:** 3 additional masked versions per sentence
- **Final training size:** approximately 4× the original training set

Environment and Dependencies

- Python 3.x
- Required libraries:
    - pandas
    - datasets
    - transformers
    - sacrebleu
    - nltk
    - numpy
    - openpyxl

Appendix C

Sample Raw Dataset and Masked Data

Sample Raw Dataset and Masked Data

The examples presented in this appendix illustrate the structure of the raw Ilocano-Tagalog parallel dataset and demonstrate the application of mask-based data augmentation to the Ilocano source sentences. Masking was applied exclusively to the source language with a probability of 0.15, while the corresponding Tagalog target sentences were retained unchanged to preserve correct source-target alignment.

The masked examples show how selected source tokens are replaced with a <mask> placeholder, introducing controlled variation in the input while maintaining the original reference translation.

The examples included here are illustrative and represent only a small subset of the full dataset used in the study. All augmented samples were generated solely from the training partition, while the validation and test sets remained unmodified to ensure fair and leakage-free evaluation.

Table C.1. Sample Raw Ilocano-Tagalog Parallel Sentences

No.	Ilocano (Source)	Tagalog (Target)
1	sumingsingpet ni juan iti trabaho iti bigat	masipag si juan sa trabaho bukas
2	aguray ti boss ditoy reception	naghihintay ang boss dito sa reception
3	agawid tayu man idiay balay ita	uwi tayo sa bahay ngayon
4	anya ti ibagbaga ni peter iti klase	ano ang sinasabi ni peter sa klase
5	ania ti nagan ti principal	ano ang pangalan ng principal
6	napudot ti init iti rabii	mainit ang araw mamaya
7	aglinis ka iti paso a tanem	linisin mo ang paso ng halaman
8	agaramid ni juan iti laruan	gumagawa si juan ng laruan
9	aguray ti team iti HR para iti orientation	naghihintay ang team sa HR para sa orientation
10	addaa orientation dagiti interns iti hr office	may orientation ang interns sa hr office

Table C.2. Sample Masked Ilocano Sentences Used for Training

No.	Original Ilocano Sentence	Masked Ilocano Sentence	Tagalog Target
1	agawid tayu man idiay balay ita	agawid tayu man <mask> <mask>	uwi tayo sa bahay ngayon
2	anya ti ibagbaga ni peter iti klase	anya <mask> ibagbaga ni peter iti klase	ano ang sinasabi ni peter sa klase
3	ania ti nagan ti principal	ania <mask> nagan ti principal	ano ang pangalan ng principal
4	agaramid ni juan iti laruan	<mask> ni juan iti laruan	gumagawa si juan ng laruan
5	aguray ti team iti HR para iti orientation	aguray ti team iti HR para iti <mask>	naghihintay ang team sa HR para sa orientation
6	addaa orientation dagiti interns iti hr office	addaa orientation dagiti interns <mask> hr office	may orientation ang interns sa hr office
7	kayatmo kadi ti agawid iti alas nuebe iti bigat	kayatmo kadi ti agawid iti alas nuebe iti <mask>	gusto mo bang umuwi ng alas nuebe ng umaga
8	kasano ti panagawid yo idiay vigan idi mayo	kasano ti panagawid yo idiay vigan <mask> mayo	kumusta ang pag-uwi ninyo mula vigan nitong mayo
9	agturog ni juan idiay eskuela	agturog ni juan <mask> eskuela	pupunta si juan sa paaralan
10	ag request man ti IT para iti password reset	ag request man ti IT para iti <mask> reset	mag request sa IT para sa password reset

Appendix D

Relevant Source Code

Relevant Code

Hardware / Runtime Environment

Model training and evaluation were implemented in Google Colab, with checkpoints and logs saved to Google Drive. GPU acceleration was enabled automatically when available using PyTorch's device selection.

Required Python Libraries

The augmented MarianMT pipeline relies on Hugging Face tooling for dataset handling and training, plus evaluation and experiment logging utilities. The environment installs the following packages:

```
!pip install -q transformers datasets sentencepiece sacrebleu wandb evaluate tensorboard
!pip install --upgrade --quiet torch transformers datasets sentencepiece
```

Key roles of the libraries:

- transformers: MarianMT model/tokenizer + Seq2SeqTrainer training loop
- datasets: dataset conversion, mapping, augmentation, and tokenization pipeline
- sentencepiece: Marian tokenizer subword segmentation dependency
- evaluate + sacrebleu: BLEU computation during evaluation
- tensorboard: training/evaluation visualization

Train-Validation-Test Split

After loading and converting the Excel data into a Hugging Face Dataset, the notebook splits the corpus into train/validation/test using a fixed random seed to keep the partitioning reproducible.

```
dataset_split = dataset.train_test_split(test_size=0.1, seed=42)
train_val_data = dataset_split["train"]
test_data = dataset_split["test"]

train_val_split = train_val_data.train_test_split(test_size=0.1, seed=42)
train_data = train_val_split["train"]
val_data = train_val_split["test"]
```

The cleaned parallel corpus was divided into training, validation, and test subsets using a two-stage random split with a fixed seed (seed = 42) to ensure reproducibility. First, 10% of the dataset was held out as a test set. The remaining 90% was then split again, with 10% allocated for validation and the remaining 81% used for training. This splitting strategy ensures that the test set remains fully unseen during training and that validation data is used solely for performance monitoring.

Mask-Based Augmentation

```
def mask_tokens(example, mask_prob=0.15, mask_token="<mask>"):
    words = str(example["ilocano"]).split()  # ensure string
    new_words = [mask_token if random.random() < mask_prob else word for word in words]
    example["ilocano_masked"] = " ".join(new_words)
    example["tagalog"] = str(example["tagalog"])
    return example
```

```
def augment_dataset(dataset, num_variants=3):
    augmented_datasets = [dataset.map(mask_tokens)]
    for _ in range(num_variants):
        masked_dataset = dataset.map(mask_tokens)
        augmented_datasets.append(masked_dataset)
    return concatenate_datasets(augmented_datasets)

train_data_augmented = augment_dataset(train_data, num_variants=3)
print("Augmented train size:", len(train_data_augmented))
```

The mask_tokens function performs data augmentation by randomly replacing a portion of words in the Ilocano source sentence with a special <mask> token, using a masking probability of 0.15. This encourages the model to learn translation behavior even when parts of the input are partially obscured, while the Tagalog target sentence remains unchanged.

The augment_dataset function applies this masking procedure multiple times to the training data, generating several masked variants of each sentence. These augmented datasets are then concatenated to form a larger and more diverse training set. Augmentation is applied only to the training split, ensuring that validation and test sets remain unmodified for fair evaluation.

Tokenization

For Augmented

```
model_inputs = tokenizer(
    examples["ilocano_masked"],
    max_length=128,
    padding="max_length",
    truncation=True
```

For Both Non-Augmented and Augmented

```
model_inputs = tokenizer(
    examples["ilocano"],
    max_length=128,
    padding="max_length",
    truncation=True
)
```

```
labels = tokenizer(
    examples["tagalog"],
    max_length=128,
    padding="max_length",
    truncation=True
```

During training, the model tokenizes masked Ilocano sentences as input, allowing it to learn translation patterns even when parts of the source sentence are obscured. For validation and testing, tokenization is applied to the original, unmasked Ilocano sentences, ensuring that evaluation reflects real translation conditions. In all splits, Tagalog sentences are tokenized as target labels, and all sequences are padded and truncated to a fixed length of 128 tokens to maintain consistent input dimensions.

Training Hyperparameters

The hyperparameters define the training, optimization, and evaluation behavior of the augmented MarianMT model. A learning rate of 5×10^{-5} is used to ensure stable fine-tuning of the pre-trained model, while weight decay (0.01) provides regularization to reduce overfitting. The model is trained for 5 epochs with a batch size of 16 for both training and evaluation.

```
training_args = Seq2SeqTrainingArguments(
    output_dir="/content/drive/MyDrive/mtTXTv9",
    eval_strategy="epoch",
    save_strategy="steps",
    save_steps=200,
    save_total_limit=5,
    learning_rate=5e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    weight_decay=0.01,
    num_train_epochs=5,
    predict_with_generate=True,    # necessary for BLEU
    generation_num_beams=5,
    logging_dir="/content/drive/MyDrive/mtTXTv9/logs",
    logging_steps=500,
    report_to="tensorboard",
    fp16 = True
)

# 4. Initialize Trainer
```

Evaluation is performed at the end of each epoch, and model checkpoints are saved every 200 training steps, with a maximum of five checkpoints retained. To enable BLEU computation during evaluation, sequence generation is activated, using beam search with 5 beams to improve translation quality. Mixed-precision training (fp16) is enabled to reduce memory usage and accelerate training on GPU hardware. Training progress and metrics are logged using TensorBoard for monitoring.

```
def compute_metrics(eval_pred):
    predictions, labels = eval_pred
    # Decode predictions
    decoded_preds = tokenizer.batch_decode(predictions, skip_special_tokens=True)
    # Replace -100 in labels as pad_token_id
    labels = np.where(labels != -100, labels, tokenizer.pad_token_id)
    decoded_labels = tokenizer.batch_decode(labels, skip_special_tokens=True)
    # Compute BLEU score
    bleu_score = bleu.compute(predictions=decoded_preds, references=[[l] for l in decoded_labels])
    return {"bleu": bleu_score["score"]}
```

Appendix E

Representative Pages from the Human Validation Materials

Representative Pages from the Human Validation Materials

This appendix presents representative excerpts from the human validation instruments used in the study, including the evaluation instructions, scoring rubrics, error annotation scheme, reference translation task, and sample Ilocano-Tagalog sentence pairs. The human evaluation component was conducted on 195 sentence pairs, while the human reference translation task was performed on a subset of 80 sentence pairs. Only selected pages from these instruments are shown here for brevity and documentation purposes.

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ENHANCING LOW-RESOURCE LANGUAGE PROCESSING: FINE-TUNING MARIANMT FOR ILOCANO - TAGALOG NEURAL MACHINE TRANSLATION

Danniel Paul B. Molleno, Dania Marie Bautista, Rebecca B. Gomez
Researchers

The primary goal of this research is to develop a new fine-tuned model version of MarianMT for translating between Ilocano and Tagalog. MarianMT is a pretrained multilingual neural machine translation (NMT) model that supports translation between many languages and can be fine-tuned for specific language pairs to improve translation quality. Our fine-tuned version aims to enhance translations for low-resource Philippine languages by training on bilingual text data and optimizing for fluency and accuracy.

This questionnaire, referred to in the study as the Evaluation Tool, is designed to assess the MarianMT model's translation performance across key linguistic dimensions: adequacy, fluency, faithfulness, and overall quality. Your professional judgment and linguistic expertise are vital in validating and refining the model's translation capability.

The MarianMT model functions as a bilingual language assistant that interprets a sentence in one language and generates a natural, contextually appropriate translation in another. Unlike direct word-for-word translation, it employs neural networks that learn grammar, semantics, and sentence structure from parallel text data, enabling it to produce more fluent and meaningful translations.

As a native speaker evaluator, your role is to critically examine the translations generated by the fine-tuned MarianMT model and determine whether they demonstrate adequacy, fluency, and faithfulness to the original Ilocano source sentences. In addition, you will perform error annotation to identify and categorize specific translation issues. Your evaluation will provide valuable insights for refining the model and ensuring its reliability for low-resource language translation tasks.

Validation Procedure

The core purpose of this validation is to assess the model's performance in accurately translating Ilocano sentences into Tagalog.

The validation process will involve the following steps:

1. **Data Provision**:
   The research team will provide the validator with a set of 195 sentence pairs. Each pair will contain an Ilocano source sentence and its corresponding Tagalog translation generated by the fine-tuned MarianMT model.

2. **Expert Validation**:
   As the validator, your knowledge is essential in determining whether each Tagalog translation accurately preserves the meaning and intent of its corresponding Ilocano source sentence. You are expected to provide an objective, professional assessment based on four criteria:

   - **Adequacy**: Measures how much of the original meaning or sense from the Ilocano sentence is preserved in the Tagalog translation.
   - **Fluency**: Evaluates how natural, grammatically correct, and well-formed the Tagalog translation is within the context of the target language.
   - **Faithfulness**: Assesses the extent to which the translation accurately conveys the intended meaning of the source text without distortion or addition.
   - **Overall Quality**: Reflects the expert's holistic judgment of the translation's overall linguistic appropriateness.

3. **Performance Marking**:
   After reviewing each translation, please rate its quality according to the Evaluation Scale below. Each sentence should be scored individually using the most appropriate rating.

Evaluation Scale

Rating	Description
5 (Excellent)	Consistently meets the criterion with high linguistic precision; no noticeable errors.
4 (Very Good)	Meets the criterion well; minor, non-critical linguistic flaws.
3 (Good)	Adequately meets the criterion; some noticeable issues but meaning remains clear.
2 (Fair)	Partially meets the criterion; translation shows several errors affecting clarity.
1 (Poor)	Fails to meet the criterion; translation is inaccurate or unclear.

4. **Error Annotation**
   In addition to numeric scoring, validators are asked to perform a qualitative error annotation for each sentence pair. This step aims to identify and categorize specific translation errors present in the model's Tagalog outputs. Validators should tick (✓) all applicable error types per sentence based on the following table:

Code	Error Type	Example
A	Mistranslation	*"agtrabaho"* → *"maglaro"* (wrong sense)
B	Omission	Missing modifiers in long sentences
C	Addition	Unnecessary words added
D	Word Order Error	Adjectives misplaced before nouns
E	Grammar / Morphology	Incorrect verb conjugations
F	Term Inconsistency	*"eskuela"* → *"paaralan"* vs *"eskuwelahan"*

This validation process will determine the degree of agreement between the MarianMT model's output and the linguistic standards expected from accurate Ilocano-Tagalog translation.

All responses will be kept strictly confidential and used solely for academic and research purposes.

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Ilocano–Tagalog Sentence Pairs for Validation

The following table presents the 195 Ilocano-Tagalog sentence pairs used in the validation process. Each Ilocano sentence was translated into Tagalog using the fine-tuned MarianMT model. These data were provided to the language expert for evaluation of translation accuracy, fluency, and faithfulness.

Pair No.	Language	Sentence Pairs
1	Ilocano	agprito ka iti bangus
	Tagalog	magprito ka ng bangus
2	Ilocano	agsukisok ni Anna iti drawer
	Tagalog	sinusuri ni anna ang drawer
3	Ilocano	napigsa ti angin iti baybay
	Tagalog	malakas ang hangin sa dagat
4	Ilocano	agluto ka iti marunggay
	Tagalog	magluto ka ng malunggay
5	Ilocano	immay kadi ni Pedro idiay sine iti mierkoles
	Tagalog	pumunta ba si Pedro sa sine noong miyerkules
6	Ilocano	manglukat ni anna iti pinto iti kwarto
	Tagalog	binuksan ni anna ang pinto sa kwarto
7	Ilocano	kasano ti biyahe yo idiay manila idi enero
	Tagalog	kumusta ang biyahe ninyo sa manila noong enero
8	Ilocano	mabalin kadi nga ibagak ti snacks iti messenger inton martes
	Tagalog	pwede ba akong sabihin sa meryenda sa messenger sa martes
9	Ilocano	kayat mo nga agjogging iti plaza iti aldaw ti domingo
	Tagalog	gusto mo bang mag joging sa plaza sa linggo
10	Ilocano	kasano ti panagluto mo iti panangaldaw
	Tagalog	kumusta ang magluto mo sa tanghalian
11	Ilocano	kasano ti trabaho ti nanangyo iti tindaan
	Tagalog	kumusta ang trabaho ng nanay ninyo sa tindahan
12	Ilocano	agpasyar tayo kadi iti parke inton bigat
	Tagalog	maglalakad tayo ba sa parke bukas
13	Ilocano	urayen nak man ditoy plaza iti lunes
	Tagalog	hintayin mo ako dito sa plaza sa lunes
14	Ilocano	agsingsingpet ka iti trabaho
	Tagalog	matapat ka ba sa trabaho
15	Ilocano	mapanak agbakasyon
	Tagalog	baka maaari mong magbakasyon
16	Ilocano	mabalin kadi nga agurnos ka iti kwarto
	Tagalog	pwede ka bang mag ayos ng kwarto
17	Ilocano	napintas ti balay iti syudad
	Tagalog	maganda ang bahay sa lungsod



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Pair No.	Language	Sentence Pairs
185	Ilocano	kasano ti rikna yo
	Tagalog	kumusta po panagkaykaykaysa ninyo
186	Ilocano	adda kadi ti plano yo a mangrugi iti grupo chat
	Tagalog	may plano ba kayo na simulan ang grupo chat
187	Ilocano	nasayaat ti agtrabaho iti laboratoryo
	Tagalog	masaya ang magtrabaho sa laboratory
188	Ilocano	agturong ka kadi idiay parke
	Tagalog	pupunta ka ba sa parke
189	Ilocano	adda kadi ti cellphone iti bag ni Pedro
	Tagalog	may cellphone ba sa bag ni Pedro
190	Ilocano	adda kadi ti itlog iti lamisa
	Tagalog	may itlog ba sa mesa
191	Ilocano	mangisagana ni anna iti merienda
	Tagalog	naghahanda si anna ng meryenda
192	Ilocano	awan pay ti pag-aprobahan ti department head
	Tagalog	wala pang approval ng department head
193	Ilocano	adda kadi ti gagayyem yo nga naka istoria iti grupo chat
	Tagalog	may kaibigan ba kayong nakausap sa group chat
194	Ilocano	adda kadi ti kanta iti radio
	Tagalog	may kanta ba sa radyo
195	Ilocano	pakisagana ti kanin ken sabaw tapno makapanganen
	Tagalog	ihanda ang kanin at sabaw para makakain

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VALIDATION TABLE

Instruction:

Performance Marking: Following your review, this scoring table is provided for you to mark the model's translation performance on a per-sentence basis.

Rating	Description
5 (Excellent)	Consistently meets the criterion with high linguistic precision; no noticeable errors.
4 (Very Good)	Meets the criterion well; minor, non-critical linguistic flaws.
3 (Good)	Adequately meets the criterion; some noticeable issues but meaning remains clear.
2 (Fair)	Partially meets the criterion; translation shows several errors affecting clarity.
1 (Poor)	Fails to meet the criterion; translation is inaccurate or unclear.

This table is designed to evaluate the quality of the Tagalog translations produced by the MarianMT model for each Ilocano source sentence.

Pair No.	Adequacy	Fluency	Faithfulness	Overall Quality
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				

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ERROR ANNOTATION TABLE

Instruction:

For each sentence pair, please **tick (✓)** the boxes corresponding to any errors observed in the model's translation output.
Multiple error types may apply to a single translation.
Refer to the table below for error type descriptions.

Code	Error Type	Example
A	Mistranslation	*"agtrabaho"* → *"maglaro"* (wrong sense)
B	Omission	Missing modifiers in long sentences
C	Addition	Unnecessary words added
D	Word Order Error	Adjectives misplaced before nouns
E	Grammar / Morphology	Incorrect verb conjugations
F	Term Inconsistency	*"eskuela"* → *"paaralan" vs "eskuwelahan"*

This table is designed to record and classify expert-identified translation errors in the Tagalog outputs produced by the MarianMT model for each Ilocano source sentence.

Pair No.	A	B	C	D	E	F	Comments (If any)
1	☐	☐	☐	☐	☐	☐	
2	☐	☐	☐	☐	☐	☐	
3	☐	☐	☐	☐	☐	☐	
4	☐	☐	☐	☐	☐	☐	
5	☐	☐	☐	☐	☐	☐	
6	☐	☐	☐	☐	☐	☐	
7	☐	☐	☐	☐	☐	☐	
8	☐	☐	☐	☐	☐	☐	
9	☐	☐	☐	☐	☐	☐	
10	☐	☐	☐	☐	☐	☐	
11	☐	☐	☐	☐	☐	☐	
12	☐	☐	☐	☐	☐	☐	
13	☐	☐	☐	☐	☐	☐	
14	☐	☐	☐	☐	☐	☐	
15	☐	☐	☐	☐	☐	☐	
16	☐	☐	☐	☐	☐	☐	

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Human Reference Translation

Instruction

Reference Translation Task:
Following the instructions below, the native speaker validator is requested to provide a Tagalog translation for each Ilocano sentence listed in the table.

Each Ilocano sentence should be translated into natural, grammatically correct, and contextually appropriate Tagalog, accurately preserving the meaning and intent of the original source sentence. The translation should reflect standard Tagalog usage and does not need to follow a word-for-word structure.

There are no correct or incorrect answers. Variations in phrasing are acceptable as long as the original meaning is clearly conveyed.

Purpose of the Table
This table is designed to collect human reference translations produced by a native speaker validator. The translations will be used exclusively as reference outputs for comparison with the Tagalog translations generated by the fine-tuned MarianMT model and will not be used for model training or fine-tuning.

No.	Ilocano Sentence
1.	mapan ka manen eskuela
2.	mangan tayon iti field trip
3.	agtugaw ka kadi ditoy silya
4.	mangisagana ni anna iti merienda
5.	adda kadi kasapulan yo

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Validation Certification

This letter serves to formally certify that I, _____, in my capacity as Native Speaker Validator, have completed the validation process for the research results presented by Dania Marie Bautista, Rebecca B. Gomez, and Danniel Paul B. Molleno for the project titled: **"ENHANCING LOW-RESOURCE LANGUAGE PROCESSING: FINE-TUNING MARIANMT FOR ILOCANO-TAGALOG NEURAL MACHINE TRANSLATION."**

The validation process, conducted on _____, focused specifically on the linguistic adequacy, fluency, and faithfulness of the translations generated by the fine-tuned MarianMT model. This involved reviewing a set of Ilocano-Tagalog sentence pairs, rating each translation according to established linguistic quality criteria, and performing an error annotation to identify and categorize specific translation errors. These combined steps ensure that the evaluation reflects the accuracy and contextual fidelity of the model's translations.

Based on this comprehensive review, I hereby certify that the results presented in the final report/documentation are consistent with the data provided and the methodology used.

Furthermore, I confirm that the model's translation performance, including both quantitative scoring and qualitative error annotation, was validated to accurately reflect the findings of the study. The translations produced are generally consistent with the linguistic standards expected for Ilocano-Tagalog language pairs within the parameters established by the research. The research team's presentation of these results represents an appropriate interpretation of the validated outputs.

_____
Signature over Printed Name
Date of Validation: _____

Appendix F

Signed Validation Certification

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Validation Certification

This letter serves to formally certify that I, ARNOLD ALNAS, in my capacity as Native Speaker Validator, have completed the validation process for the research results presented by Dania Marie Bautista, Rebecca B. Gomez, and Danniel Paul B. Molleno for the project titled: **"ENHANCING LOW-RESOURCE LANGUAGE PROCESSING: FINE-TUNING MARIANMT FOR ILOCANO-TAGALOG NEURAL MACHINE TRANSLATION."**

The validation process, conducted on DEC 10 2025, focused specifically on the linguistic adequacy, fluency, and faithfulness of the translations generated by the fine-tuned MarianMT model. This involved reviewing a set of Ilocano-Tagalog sentence pairs, rating each translation according to established linguistic quality criteria, and separately producing Tagalog translations from a designated subset of Ilocano source sentences, which were used as the human reference translations for the study. These combined steps ensure that the evaluation reflects the accuracy and contextual fidelity of the model's translations.

Based on this comprehensive review, I hereby certify that the results presented in the final report/documentation are consistent with the data provided and the methodology used.

Furthermore, I confirm that the model's translation performance, including quantitative scoring and qualitative linguistic evaluation, was validated to accurately reflect the findings of the study. The human reference translations produced from the Ilocano source sentences were used as the basis for assessing model outputs, ensuring consistency between the evaluation criteria and the reference material. The translations produced are generally consistent with the linguistic standards expected for Ilocano-Tagalog language pairs within the parameters established by the research. The research team's presentation of these results represents an appropriate interpretation of the validated outputs.

ARNOLD ALNAS
Signature over Printed Name

Date of Validation: DEC 10 2025

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Validation Certification

This letter serves to formally certify that I, VICTORIA B. TOBIAS, in my capacity as Native Speaker Validator, have completed the validation process for the research results presented by Dania Marie Bautista, Rebecca B. Gomez, and Danniel Paul B. Molleno for the project titled: **"ENHANCING LOW-RESOURCE LANGUAGE PROCESSING: FINE-TUNING MARIANMT FOR ILOCANO-TAGALOG NEURAL MACHINE TRANSLATION."**

The validation process, conducted on Dec 12, 2025, focused specifically on the linguistic adequacy, fluency, and faithfulness of the translations generated by the fine-tuned MarianMT model. This involved reviewing a set of Ilocano-Tagalog sentence pairs, rating each translation according to established linguistic quality criteria, and performing an error annotation to identify and categorize specific translation errors. These combined steps ensure that the evaluation reflects the accuracy and contextual fidelity of the model's translations.

Based on this comprehensive review, I hereby certify that the results presented in the final report/documentation are consistent with the data provided and the methodology used.

Furthermore, I confirm that the model's translation performance, including both quantitative scoring and qualitative error annotation, was validated to accurately reflect the findings of the study. The translations produced are generally consistent with the linguistic standards expected for Ilocano-Tagalog language pairs within the parameters established by the research. The research team's presentation of these results represents an appropriate interpretation of the validated outputs.

V Tobias
VICTORIA B. TOBIAS
Signature over Printed Name
Date of Validation: Dec 12, 2025

Appendix G

Sample Input and Output of the Ilocano-Tagalog Translation System

Sample Input and Output of the Ilocano-Tagalog Translation System

This appendix presents selected examples of Ilocano input sentences and their corresponding Tagalog translations generated by the proposed Ilocano-Tagalog MarianMT translation system. For reference and qualitative comparison, outputs from the non-augmented baseline model, Google Translate, and ChatGPT are also included alongside the human reference translations.

The examples are intended to illustrate typical translation behavior across different systems and sentence structures. All outputs are shown without post-editing to reflect the natural responses of each translation system. No quantitative evaluation or error analysis is conducted in this appendix, as performance assessment is discussed in the main results chapter.

Table E.1 Sample Ilocano Input Sentences and Corresponding Tagalog Outputs

No.	Ilocano Source Text	Human Reference (Target Text)	Model Output	No-Augment Model	Google Translate	ChatGPT
1	kayat mo kadi ti agluto	gusto mo bang magluto	gusto mo bang magluto	gusto mo bang magluto	gusto mo bang magluto	gusto mo bang magluto
2	adda ka kadi idiay klase	nasa klase ka ba	nasa klase ka ba	nasa klase ka ba	nasa klase ka ba	nandiyan ka ba sa klase
3	ania ti nagan ti presidente	ano ang pangalan ng presidente	ano ang pangalan ng empleyado	ano ang pangalan ng presidente	ano ang pangalan ng pangulo	ano ang pangalan ng presidente
4	sagrapem dagiti hugasan a plato	ipunin ang mga platong huhugasan	ipunin ang mga platong huhugasan	ipunin ang mga platong huhugasan	tamasahin ang mga hugasan na pinggan	ipunin mo ang mga huhugasan na plato
5	kayat yo kadi agbasa	nais po bang magbasa	nais po bang mag aral	nais po bang magbasa	ikalat ito nang hindi nahugasan	gusto niyo bang magbasa
6	kayat yo kadi agturong	nais po bang pumunta	nais po bang pumunta	nais po bang pumunta	gusto mo bang lumiko	gusto niyo bang pumunta
7	nasayaat ti agtrabaho iti laboratory	masaya ang magtrabaho sa laboratory	masaya ang magtrabaho sa laboratory	masaya ang paggawa ng laboratory	ang pagtatrabaho sa laboratoryo ay mabuti	maganda ang pagtatrabaho sa laboratoryo
8	nasayaat ti panagkaykay sa iti pamilya	maganda ang pagkakais	maganda ang	masaya ang	magandang pagkakaisa ng pamilya	maganda ang

		a sa pamilya	samahan sa pamilya	pagkain sa pamilya		pagkakaisa ng pamilya
9	anya ti ibagbaga ni juan	ano ang sinasabi ni juan	ano ang sinasabi ni juan	ano ang sinasabi ni juan	anong pinagsasabi ni juan	ano ang sinasabi ni juan
10	mangdaldalus ni anna iti kwarto	naglilinis si anna ng kwarto	naglilinis si anna ng kwarto	naglilinis si anna ng kwarto	naglilinis ng kwarto si anna	naglilinis si anna sa kwarto
11	adda sabong iti kaldero	may sabaw sa kaldero	may sabaw sa kaldero	may bulaklak sa kaldero	may mga bulaklak sa palayok	may sabon sa kaldero
12	agtugaw ni maria iti balkonahe	nakaupo si maria sa balkonahe	nakaupo si maria sa balkonahe	nakaupo si maria sa balkonahe	nakaupo si maria sa balcony	nakaupo si maria sa balkonahe
13	immaykadi ni lolo idiay balay	pumunta ba si lolo sa bahay	pumunta ba si lolo sa bahay	pumunta ba si lolo sa bahay	umuwi ba si lolo	dumating ba si lolo sa bahay
14	ania ti parokya yo	ano po parokya ninyo	ano po parokya ninyo	ano po parokya ninyo	ito ang iyong parokya	ano ang parokya ninyo
15	kayat yo kadi agsapul	nais po bang humingi	nais po bang humingi	nais po bang humingi	gusto mo bang maghanap	gusto niyo bang maghanap
16	napintas ti tan-aw iti baybay	maganda ang tanawin sa dagat	maganda ang tanawin sa dagat	maganda ang tanawin sa dagat	ang ganda ng view sa dagat	maganda ang tanawin sa dagat
17	agsagana ka iti bigas	magsaing ka ng kanin	maghanda ka ng bigas	naghahanda ka ng bigas	ihanda ang sarili sa kanin	maghanda ka ng bigas
18	manginana dagiti ubbing iti dalan	umiingay ang mga bata sa kalsada	umiingay ang mga bata sa kalsada	umiingay ang mga bata sa kalsada	nagpapahinga ang mga bata sa daan	naglalaro ang mga bata sa daan
19	pakilinis ti agdan iti veranda	pakilinis ng hagdan sa veranda	pakilinis ng hagdan sa veranda	pakilinis ng hagdan sa veranda	pakilinis ang hagdan sa veranda	pakilinis ang hagdan sa veranda
20	kayat yo kadi agserbi	nais po bang maglingkod	nais po bang maglingkod	nais po bang maglingkod	gusto mo bang maglingkod	gusto niyo bang maglingkod

Appendix J

Curriculum Vitae



Dania Marie Bautista

Graduating Student

+63 995 643 3167

bdania871@gmail.com

Los Baños, Laguna

Professional Summary

A Computer Science student with a passion for design and technology. Experienced in customer service, teamwork, and problem-solving skills. Seeking opportunities to apply technical knowledge and interpersonal abilities in a dynamic work environment.

Education

Bachelor's Degree in Computer Science

Laguna State Polytechnic University
Los Baños Laguna
Expected Graduation: 2026
2022 – Present

General Academic Strand (GAS)

Los Baños Bayog Senior High School
Los Baños, Laguna
2021 – 2022

Trainings & Seminars Attended

Role of Technology in Sustainable Development
February 2025

Securing the Future: Understanding the Fundamentals of Cyber Security
December 2022

Academic Works

Low-Resource Language Neural Machine Translation
Undergraduate Thesis, 2026

- Fine-tuned a MarianMT neural machine translation model for Ilocano-Tagalog low-resource language translation.

College Inquiry Chatbot
Software Engineering, 2025

- Created an AI-driven college inquiry chatbot that delivers instant academic and admissions information through natural language interaction..

SafePaws
Robotics, 2025

- Developed a web-based pet tracking system using an ESP32 and NEO-6M GPS module with real-time geofence alerts, integrating machine learning (KNN) to analyze pet movement and behavior patterns

Skills

- Figma
- Canva
- Microsoft Office
- Problem Solving
- English and Filipino Communication
- Time Management
- Knowledge Acquisition

Work Experience

Service Crew
Jollibee Foods Corporation (JFC)
Jollibee Olivarez Plaza Los Baños
April 2022 - September 2022



Rebecca Gomez

Graduating Student

+63 975 156 8124

rebeccabgomez@gmail.com

Los Baños, Laguna

Professional Summary

Computer Science undergraduate with hands-on experience in Python and AI-focused academic projects, including neural machine translation and resume–job matching systems. Possesses a strong foundation in problem-solving, data organization, and documentation, with a keen interest in applying technical skills to real-world IT solutions.

Education

Bachelor's Degree in Computer Science

Laguna State Polytechnic University
Los Baños Laguna
Expected Graduation: 2026
2022 – Present

Bachelor's Degree in Nutrition

University of the Philippines
Los Baños Laguna
Undergraduate Studies
2006 – 2009

Trainings & Seminars Attended

Role of Technology in Sustainable Development
February 2025

Securing the Future: Understanding the Fundamentals of Cyber Security
December 2022

Academic Works

Low-Resource Language Neural Machine Translation
Undergraduate Thesis, 2026

- Fine-tuned a MarianMT neural machine translation model for Ilocano-Tagalog low-resource language translation.

ResuMatch
Software Engineering, 2025

- Developed an AI-based resume–job matching system using Python and NLP with an HTML interface.

PushUp Bot
Robotics, 2025

- Built and programmed a push-up robot using Arduino Uno with basic hardware integration and control logic.

Skills

- Python
- HTML
- Microsoft Office
- Adobe Photoshop
- Problem Solving
- Written & Verbal Communication
- Time Management & Prioritization
- Adaptability & Learning Agility



Danniel Paul Molleno

Graduating Student

+63 9673644304

dannielpaulmolleno.15@gmail.com

Bayog, Los Baños, Laguna

Professional Summary

Aspiring IT professional with skills in Python, HTML, and C#, along with strong proficiency in Microsoft Office. I have experience in problem solving, basic animation, and effective sales-related interactions. I am highly adaptable, and motivated to continuously develop my skills in the field of information technology.

Education

Bachelor's Degree in Computer Science

Laguna State Polytechnic University
Los Baños Laguna
Expected Graduation: 2026
2022 – Present

Trainings & Seminars Attended

Securing the Future: Understanding the Fundamentals of Cyber Security
December 2022

Work Experience

Pru Life UK
Financial Advisor
2025 - Present

Personal Collection
Sales Representative
2024 - Present

Avon Philippines
Sales Representative
2023 - Present

Academic Works

Low-Resource Language Neural Machine Translation
Undergraduate Thesis, 2026

- Fine-tuned a MarianMT neural machine translation model for Ilocano-Tagalog low-resource language translation.

ResuMatch
Software Engineering, 2025

- Developed an AI-based resume–job matching system using Python and NLP with an HTML interface.

WordBridge
Thesis Project, 2025

- a Flutter-based learning app designed for high school students to strengthen math vocabulary through interactive quizzes and guided lessons, making key mathematical terms easier to understand and remember.

Skills

- Python
- HTML
- C Sharp
- Flutter App
- Microsoft Office
- Animation
- Problem Solving
- Sales Talking
- Adaptability & Learning Agility

Appendix H

Grammarly and Turnitin Report

GRAMMARLY REPORT

MARIANMT CHAPTERS 1-3

by Rebecca Gomez

General metrics

66,420	8,694	451	34 min 46 sec	1 hr 6 min
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Measures average word length

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19.3

words per sentence

Measures average sentence length

MARIANMT CHAPTERS 4-5

by Rebecca Gomez

General metrics

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Rare Words

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rare words

Word Length

Measures average word length

5.9

characters per word

Sentence Length

Measures average sentence length

9.7

words per sentence

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Appendix I

Publishable Paper (IEEE Format)

ENHANCING LOW-RESOURCE LANGUAGE PROCESSING: FINE-TUNING MARIANMT FOR ILOCANO-TAGALOG NEURAL MACHINE TRANSLATION

REBECCA B. GOMEZ, DANNIEL PAUL B. MOLLENO, DANIA MARIE BAUTISTA

College of Computer Studies (CCS), Laguna State Polytechnic University, Los Baños, Laguna, Philippines

Article Info

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Keywords: *Ilocano–Tagalog translation, neural machine translation, MarianMT, low-resource languages, data augmentation, bootstrap resampling*

ABSTRACT

A controlled neural machine translation framework for Ilocano–Tagalog is presented to address data scarcity in low-resource Philippine languages. Two MarianMT-based models were fine-tuned: a non-augmented baseline and a mask-augmented variant. From 42,273 raw entries, 10,038 high-quality parallel sentence pairs were retained after cleaning and alignment. Mask-based augmentation expanded the training set from 8,130 to 32,520 pairs, increasing lexical and syntactic diversity. Translation quality was evaluated using BLEU, ChrF, METEOR, human assessment, and paired bootstrap resampling. The augmented model significantly outperformed the baseline (BLEU = 88.24, ChrF = 92.86, METEOR = 0.9284; $p < 0.001$) and exhibited fewer human-annotated errors. Although absolute metric values are high, they reflect controlled dataset characteristics, including limited syntactic variability and close linguistic similarity between Ilocano and Tagalog. Accordingly, emphasis is placed on relative performance gains rather than absolute scores. Results demonstrate that mask-based augmentation improves translation accuracy and consistency in low-resource NMT. Future work will explore additional data sources, augmentation strategies, and multilingual transfer.

Corresponding Author:

Name: Rebecca B. Gomez, Danniel Paul B. Molleno, Dania Marie Bautista

College of Computer Studies,

Laguna State Polytechnic University,

Malinta, Los Baños, Laguna, Philippines 4030.

Email: rebeccabgomez@gmail.com, dannielpaulmolleno.15@gmail.com, bdania871@gmail.com

1. INTRODUCTION

Digital technologies have reshaped language from a tool for interpersonal communication into a gateway for accessing information, participating in civic life, and engaging in public discourse. In multilingual settings, the language a person speaks can determine whether they can benefit from digital services and innovations, with direct implications for inclusion and equity in the digital sphere [10]. This issue is especially salient in the Philippines, a highly multilingual country with over 180 languages spoken across the archipelago [5]. Despite this diversity, language technologies have historically prioritized dominant languages, particularly English and Tagalog, resulting in uneven digital visibility across Philippine languages [10].

Tagalog has received comparatively greater representation in multilingual NLP resources and translation research, supported by large-scale multilingual pretraining and open translation infrastructures [6], [13]. In contrast, Ilocano, spoken by millions and actively used in daily communication, remains underrepresented in computational tools, curated corpora, and translation resources [12]. Available Ilocano text is also frequently shaped by code-switching and mixed-language usage, which can complicate the creation

of clean, well-aligned training data for machine translation [20]. These gaps contribute to language-based inequality: limited Ilocano language technology can constrain access to digital content, education-related materials, and public information, reinforcing regional and socio-economic disparities [10]. Recent evidence further suggests shifting language preferences among younger speakers in educational and online contexts, underscoring the urgency of technological support for sustained language use and intergenerational transmission [12].

Global and institutional initiatives increasingly emphasize inclusive language technologies, including UNESCO's International Decade of Indigenous Languages (2022–2032) and large-scale efforts to expand machine translation coverage for underserved languages [18], [14]. However, multilingual machine translation systems often depend on large, balanced datasets and substantial compute, conditions that rarely hold for low-resource languages with limited, uneven, or domain-specific corpora [8]. As a result, pre-trained multilingual models may underperform for under-resourced languages when language-specific data are insufficient to capture lexical diversity, morphological variation, and context-dependent meaning [8].

A practical response is targeted adaptation through fine-tuning, where a pre-trained neural machine translation model is continued on a smaller, language-pair-specific corpus to better match the linguistic properties of the target setting [8]. MarianMT, implemented in the Marian framework, is designed for efficient neural machine translation and supports research-scale fine-tuning under limited resources [9]. Accordingly, this work focuses on fine-tuning MarianMT for Ilocano-to-Tagalog translation in a controlled low-resource setup and analyzing model behavior under systematic evaluation. Specifically, the study (1) constructs and preprocesses an Ilocano–Tagalog parallel corpus with cleaning, normalization, and controlled mask-based augmentation [3], [4]; (2) fine-tunes baseline and augmented MarianMT variants to examine the effect of augmentation on learning [9]; and (3) evaluates translation outputs using automatic metrics and human-assisted assessment, supported by ablation and statistical testing [1], [7], [11], [15], [16].

This work is limited to unidirectional, text-based translation (Ilocano-to-Tagalog) using a single MarianMT-based architecture. It does not explore alternative architectures, speech, or multimodal translation, or external linguistic resources beyond those inherent to the pre-trained model. While constrained by corpus size and domain coverage typical of low-resource settings [8], the controlled design provides a reproducible foundation for building and evaluating translation systems for underrepresented Philippine languages and for strengthening evaluation practices in low-resource machine translation.

2. Related Literature

This chapter reviews theoretical and empirical foundations for developing an Ilocano–Tagalog neural machine translation (NMT) system in a low-resource setting. Prior studies associate linguistic context, model architecture, fine-tuning strategies, and evaluation design with persistent challenges in low-resource translation, particularly within neural and transfer-based frameworks [8]. Despite the Philippines' linguistic diversity, limited digital corpora and computational resources continue to restrict data-driven translation research for many local languages, including Ilocano [5], [12], motivating efficient learning strategies under constrained data conditions [8].

Machine translation has evolved from rule-based and statistical approaches toward neural systems, with transformer architectures enabling effective sequence modeling through attention and parallel computation [19]. Multilingual transformer-based models support transfer learning for under-resourced languages by leveraging cross-lingual representations learned from higher-resource data [6], [13]. Fine-tuning remains essential for adapting pretrained models to specific language pairs [8], with MarianMT providing bilingual checkpoints and research-oriented workflows suitable for controlled fine-tuning experiments [9]. Data augmentation mitigates data scarcity by increasing lexical coverage while preserving semantic consistency [3], and subword tokenization addresses vocabulary sparsity and rare word forms [17].

Reliable evaluation in low-resource NMT requires attention to overfitting and generalization [8] and is commonly strengthened through combined automatic and human assessment. Automatic metrics such as BLEU, ChrF, and METEOR measure n-gram overlap, character-level similarity, and flexible unigram matching [1], [15], [16], with statistical significance often assessed using paired bootstrap resampling [11]. Human evaluation typically measures adequacy, fluency, and faithfulness, though subjectivity necessitates careful protocol design [7]. Likert-scale ratings provide a standardized approach for aggregating evaluator judgments [2].

Guided by these foundations, this study adopts a three-phase framework comprising data preparation, model training, and evaluation. An Ilocano–Tagalog parallel corpus was compiled, cleaned, aligned, and tokenized using subword segmentation [4], [17]. Mask-based augmentation was applied only to

training data to enable controlled comparison with a non-augmented baseline [3]. MarianMT fine-tuning and layered evaluation combining automatic metrics, bootstrap testing, and human assessment were used to examine translation behavior under realistic low-resource conditions [1], [2], [7], [9], [11], contributing methodological insights for underrepresented Philippine languages [10], [14], [18].

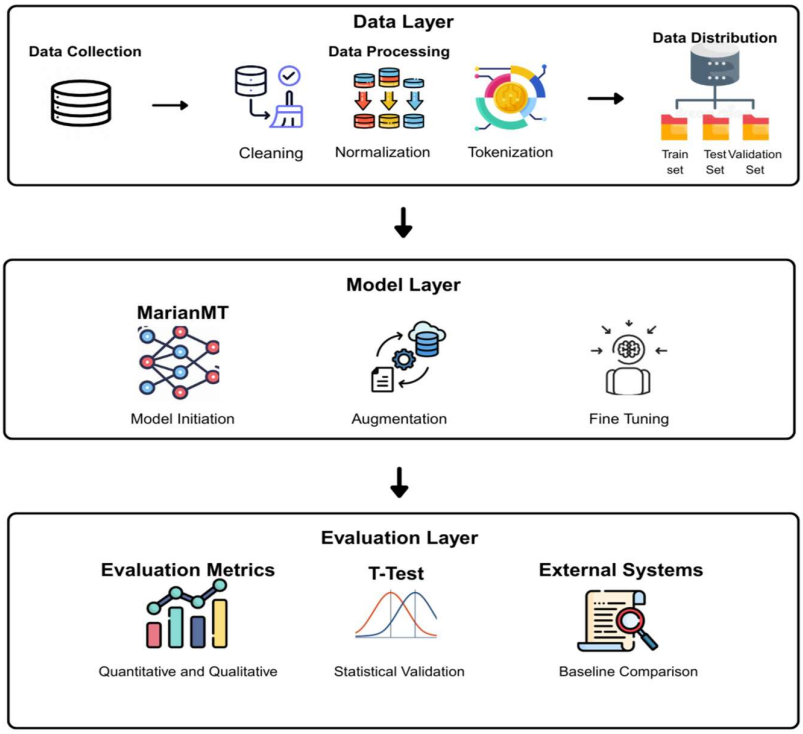


Figure 1. Conceptual Framework of the Study

3. METHODOLOGY

3.1 Research Design & Model Development

This study employed a quantitative experimental research design to evaluate the effect of source-side mask-based data augmentation on Ilocano–Tagalog neural machine translation using the MarianMT framework. Experiments were conducted in a controlled digital environment to isolate the impact of augmentation by holding model architecture, hyperparameters, and optimization settings constant. The independent variable was the application of mask-based data augmentation during training, while the dependent variables consisted of translation quality scores obtained from automatic evaluation metrics and supported by human-assisted assessment.

Two experimental configurations were implemented: a non-augmented baseline MarianMT model and an augmented MarianMT model trained with mask-based data augmentation. Both models shared identical Transformer architectures, tokenization schemes, and training configurations to ensure comparability and to attribute observed differences specifically to augmentation. Quantitative evaluation was performed on a strictly held-out test set using BLEU, ChrF, and METEOR, with paired bootstrap resampling applied to assess the statistical reliability of observed performance differences. Subset-based analyses, including human-produced reference translations, Likert-scale human evaluation, and manual error annotation, were conducted as supplementary diagnostic evaluations rather than for population-level generalization.

The Ilocano–Tagalog translation system followed a linear workflow separating data preparation, model training, output generation, and evaluation. An aligned parallel corpus was processed through cleaning, normalization, and subword tokenization using the MarianMT SentencePiece tokenizer, after which the data were partitioned into training, validation, and test sets. Mask-based augmentation was applied exclusively to the training split. The fine-tuning stage adapted a

pre-trained MarianMT model to the target language pair, and translations were generated for unseen inputs prior to independent evaluation and result analysis.

Model development focused on controlled fine-tuning of MarianMT under low-resource conditions. The baseline model was trained on 8,130 sentence pairs, while the augmented model was trained on an expanded set of 32,520 sentence pairs generated through source-side masking. Training behavior was monitored using validation loss and automatic metrics to maintain stable optimization and reduce overfitting risk. In addition to full test-set evaluation, system-level comparisons were conducted using an 80-sentence subset with human translations as references, and a separate 195-sentence subset was used for human-assisted evaluation and error annotation. Overall, this integrated research design supports systematic investigation of data augmentation effects on Ilocano–Tagalog neural machine translation under controlled low-resource conditions.

3.2 Model Evaluation

Model performance was evaluated using a combination of automatic metrics, statistical significance testing, and human-assisted assessment. Automatic evaluation employed BLEU, ChrF, and METEOR to measure surface-level overlap, character-level similarity, and semantic correspondence with reference translations. BLEU scores were computed using the SacreBLEU implementation provided by the Hugging Face Evaluate library to ensure standardized and reproducible scoring.

ChrF was used as a complementary character-based metric suitable for morphologically rich languages and less sensitive to tokenization differences. METEOR was employed as an additional metric incorporating unigram alignment and recall weighting to allow flexible matching for semantically equivalent translations. METEOR scores were computed using the NLTK implementation.

To assess whether observed performance differences were statistically reliable, paired bootstrap resampling was applied to sentence-level BLEU scores over 10,000 iterations, producing empirical confidence intervals and p-values without assuming normal score distributions.

Human evaluation was conducted as a complementary qualitative assessment. A randomly sampled subset of 195 sentence pairs from the held-out test set was evaluated using 5-point Likert-scale ratings for adequacy, fluency, and faithfulness by two independent bilingual evaluators. Evaluations were performed without access to model identity, and mean scores were computed across evaluators. Manual error annotation was performed on the same subset by a single evaluator to identify recurring translation error categories. Separately, a subset of 80 Ilocano sentences was manually translated into Tagalog to serve as a human reference for automatic metric-based comparison across the augmented model, non-augmented model, Google Translate, and ChatGPT. The subset size was selected to balance feasibility and representativeness.

3.3 Ethical Considerations

This study used only publicly available, non-sensitive textual data obtained with proper attribution, and no personal or identifiable information was collected. Data handling followed established ethical research standards emphasizing transparency and reproducibility. Human participation was limited to voluntary academic evaluation of Ilocano–Tagalog translation outputs, with informed consent and the right to withdraw. All evaluation data were anonymized and reported only in aggregated form, in compliance with the Data Privacy Act of 2012 (Republic Act No. 10173). The system was explicitly framed as a research prototype, acknowledging potential inaccuracies in low-resource settings and requiring human oversight for any practical use.

4. RESULTS AND DISCUSSION

4.1 Data Sources, Corpus Construction, and Augmentation

The Ilocano–Tagalog parallel corpus used in this study was constructed by aggregating sentence data from multiple publicly available repositories and researcher-generated resources. Sources included GitHub-based translation projects (JakeCob and sheyn018), the OPUS No Language Left Behind (NLLB) corpus, the e-SALIN dataset hosted on Hugging Face, and researcher-constructed sentence pairs designed to improve coverage of conversational and general-domain usage. These sources yielded a total of 42,273 raw sentence entries prior to preprocessing.

Systematic data cleaning and preprocessing were applied to transform the merged corpus into a sentence-level parallel dataset suitable for neural machine translation. Cleaning stages included removal of empty or punctuation-only entries, duplicate sentences, extreme-length sentences, and poorly aligned sentence pairs based on a length ratio threshold. After preprocessing and alignment validation, 10,038 high-quality parallel sentence pairs were retained. Text normalization and subword tokenization were performed using the MarianMT SentencePiece tokenizer to ensure consistent representation and mitigate vocabulary sparsity.

Table 4.1.1. Final Dataset Distribution and Train-Validation-Test Split

Dataset Partition	Number of Sentence Pairs	Percentage (%)	Purpose
Training	8,130	81.0%	Model fine-tuning and data augmentation
Validation	904	9.0%	Model monitoring and parameter evaluation
Test	1,004	10.0%	Final performance evaluation
Total	10,038	100%	-

To ensure leakage-free evaluation, the finalized corpus was split prior to augmentation using a two-step holdout strategy, yielding 8,130 training pairs, 904 validation pairs, and 1,004 test pairs (approximately 81%/9%/10%). Strict separation between splits was enforced throughout all experimental stages, and sentence-level checks were applied to confirm the absence of duplicate or near-duplicate entries across partitions.

Mask-based data augmentation was applied exclusively to the training set to address data scarcity. Ilocano source sentences were tokenized at the word level, and each token was independently replaced with a <mask> token with a probability of 15%, while the Tagalog target sentence remained unchanged. Multiple masked variants were generated per sentence, expanding the training set from 8,130 to 32,520 sentence pairs. Validation and test sets were kept unchanged. A short ablation analysis comparing masking rates of 10%, 15%, and 30% guided the selection of the final configuration.

Table 4.1.2. Summary of Dataset Distribution Before and After Augmentation

Dataset Split	Before Augmentation	After Augmentation
Training	8,130	32,520
Validation	904	904
Test	1,004	1,004
Total	10,038	34,428

The final corpus consists primarily of short to medium-length sentences with strong source–target length correlation and a Zipfian word frequency distribution. While this structure supports sentence-level translation modeling, the limited representation of longer and syntactically complex sentences constrains generalization beyond the dataset conditions. Overall, the dataset preparation and augmentation strategy provide a controlled and reproducible foundation for low-resource Ilocano-Tagalog neural machine translation experiments.

4.2 Evaluation Results and Analysis

Automatic evaluation was conducted on a held-out test set of 1,004 Ilocano–Tagalog sentence pairs using BLEU, ChrF, and METEOR to measure word-level overlap, character-level similarity, and semantic correspondence with reference translations. The mask-augmented MarianMT model achieved higher scores across all metrics (BLEU = 88.24, ChrF = 92.86, METEOR = 0.9284) than the non-augmented baseline (BLEU = 81.93, ChrF = 89.25, METEOR = 0.8884). These gains indicate improved lexical alignment, character consistency, and semantic matching. Although absolute scores are high relative to many low-resource benchmarks, they reflect the controlled dataset characteristics, including limited syntactic variability and restrained idiomatic usage. Accordingly, emphasis is placed on relative performance differences rather than absolute metric magnitude.

Output-level comparison confirms consistent improvements attributable to mask-based augmentation, with increases of +6.31 BLEU, +3.61 ChrF, and +0.0400 METEOR over the non-augmented model. Sentence-length-based analysis further shows that the augmented model maintains higher performance across short, medium, and long sentence categories, indicating that augmentation benefits persist across varying input lengths rather than being confined to specific sentence types.

Table 4.2.1 Output-level comparison of augmented and non-augmented models.

Metric	Augmented	Non-Augmented	Δ
BLEU	88.24	81.93	+6.31
ChrF	92.86	89.25	+3.61
METEOR	0.9284	0.8884	+0.0400

System-level comparison using a subset of 80 Ilocano sentences with human-produced Tagalog references shows that both MarianMT variants outperform Google Translate and ChatGPT across all automatic metrics. Targeted fine-tuning on an Ilocano-Tagalog parallel corpus enables closer alignment with human reference translations than general-purpose multilingual systems, which rely on indirect transfer and produce more paraphrastic outputs that are penalized by surface-based metrics.

Statistical significance testing using paired bootstrap resampling on sentence-level BLEU scores confirms that observed improvements are not attributable to random variation. The augmented model demonstrates a mean BLEU improvement of 12.47 points over the baseline, with a 95% confidence interval of [10.56, 14.41] and a one-

sided p-value below 0.001, providing strong statistical evidence of consistent performance gains.

Table 4.2.2. Inter-Rater Agreement Results Using Weighted Cohen's κ

Model Variant	Dimension	κ (Weighted)	N
Augmented	Adequacy	0.204	195
	Fluency	0.296	195
	Faithfulness	0.230	195
	Overall	0.254	195
Non-Augmented	Adequacy	0.338	195
	Fluency	0.105	195
	Faithfulness	0.243	195
	Overall	0.242	195

Human evaluation on 195 sentence pairs shows high perceived translation quality for both models, with mean ratings exceeding 4.4 out of 5 across adequacy, fluency, faithfulness, and overall quality. The non-augmented model exhibits slightly higher mean fluency and overall scores with lower variance, indicating more uniform surface realization. The augmented model shows marginally lower means with higher variability, consistent with increased lexical and paraphrastic diversity. Inter-rater agreement, measured using weighted Cohen's κ, ranges from low to fair across evaluation dimensions for both models, reflecting inherent subjectivity in sentence-level semantic judgment.

Error annotation analysis reveals fewer mistranslations and grammatical errors in the augmented model compared with the non-augmented baseline, while omission rates remain comparable. Addition and word-order errors occur only in the non-augmented variant. These findings align with automatic metric trends and indicate that mask-based augmentation contributes to improved semantic accuracy and grammatical consistency without degrading perceived translation quality.

4.3 Model Validation

Model validation assessed training stability, convergence, and sentence-level generalization. Across five epochs, both MarianMT variants showed stable optimization, with decreasing loss and increasing BLEU scores. The augmented model converged earlier and at a higher level than the non-augmented baseline under identical training settings, indicating more effective learning.

Generalization was evaluated using a strict no-overlap condition that reduced the test set from 1,004 to 692 sentence pairs. Both models showed moderate drops in BLEU, ChrF, and METEOR, with the augmented model exhibiting a smaller BLEU reduction (≈3.2 points) than the baseline (≈3.9 points). These results indicate stable training behavior and improved robustness of the augmented model when sentence reuse is removed.

5. SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

5.1 Summary of the Study

Low-resource neural machine translation is constrained by limited parallel data and narrow domain coverage. This study examined the effect of source-side mask-based data augmentation on Ilocano–Tagalog translation using two MarianMT models trained under identical settings: a non-augmented baseline and an augmented model. From an initial corpus of 42,273 sentence pairs, systematic cleaning and validation retained 10,038 high-quality pairs, primarily short and conversational. The data were split into training, validation, and test sets (81–9–10), with augmentation applied only to the training set, expanding it from 8,130 to 32,520 sentence pairs.

Automatic evaluation on a held-out test set of 1,004 sentences shows higher scores for the augmented model (BLEU 88.24, ChrF 92.86, METEOR 0.9284) relative to the baseline, with improvements observed across short, medium, and long sentences. Paired bootstrap resampling reports a mean BLEU difference of 12.47 points (95% CI [10.56, 14.41], $p < 0.001$). Under a strict no-overlap condition reducing the test set to 692 sentences, both models show moderate score decreases, indicating increased evaluation difficulty rather than sentence memorization. Human evaluation on 195 sentences reports mean ratings above 4.4 for both models, with slightly higher fluency for the baseline and greater lexical variation for the augmented model. Overall, results indicate that moderate mask-based augmentation improves translation performance within the controlled dataset and evaluation conditions..

5.2 Conclusions

This study examined the effect of source-side mask-based data augmentation on Ilocano–Tagalog neural machine translation using MarianMT under controlled conditions. From 42,273 raw sentence pairs, systematic cleaning retained 10,038 parallel pairs, primarily short to medium-length and structurally aligned. Mask-based augmentation was applied only to the training split, expanding it from 8,130 to 32,520 sentence pairs while preserving validation and test separation.

MarianMT models trained with and without augmentation under identical settings showed stable convergence across five epochs, with the augmented model reaching higher BLEU scores earlier and converging at a higher level. Ablation analysis indicates that moderate masking outperforms more aggressive masking, suggesting that excessive masking can remove critical lexical information in low-resource settings.

Evaluation on a held-out test set of 1,004 sentence pairs shows consistent improvements for the augmented model across BLEU, ChrF, and METEOR, with gains observed across sentence-length categories. Under a strict no-overlap condition, both models maintain strong performance despite moderate score reductions, indicating that improvements are not driven by sentence memorization. Paired bootstrap resampling confirms statistically significant BLEU differences. Human evaluation reports high quality for both models, with greater lexical variability and fewer semantic errors observed in the augmented model. Overall, the results support cautious conclusions that controlled mask-based augmentation can improve Ilocano–Tagalog translation within the constraints of the dataset and experimental design.n.

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